



GE HEALTHCARE

Technical Publications

GE Part Number 5343114
Revision 9

TEAL PDU Module in XGD2 Cabinet

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Operating Documentation

Table of Contents

IMPORTANT SAFETY INSTRUCTIONS	2
1- INTRODUCTION	8
1.1 General Description	8
1.2 Scope	9
2- SPECIFICATIONS	9
2.1 Electrical Specifications	9
2.2 Environmental Specifications	10
2.3 Mechanical Specifications	10
3- INSTALLATION AND OPERATION	11
3.1 Installation	11
3.1.1 Input Voltage Selection	12
3.1.2 Connection	13
3.2 Operation	14
3.2.1 Main Input Circuit Breaker	14
3.2.2 Power Off	14
3.2.3 EMO Reset	14
3.2.4 TVSS Indicators	14
3.2.5 Phase Indicators	14
3.2.6 Input / Output Voltage Test Points	15
3.3 Closing	16
3.3.1 Closing Unit Front panel	16
4- FIELD REPLACEABLE UNITS (FRU'S)	16
4.1 FRU Installation	17
4.1.1 Replacement of the TVSS Module	18
4.1.2 Removal of Front Panel to Access Additional FRUs	18
4.1.3 Replacement of the Control PC Board	21
4.1.4 Replacement of Contactors (CON1 or CON2)	27
4.1.5 Replacement of Fans (FAN1 or FAN2)	28
4.1.5.1 Replacement of Fans (FAN1 or FAN2) for units PN 6550012-06 Rev A3 and Prior ..	29
4.1.5.2 Replacement of Fans (FAN1 or FAN2) for units PN 6550012-06 Rev A4 and After ..	31
4.1.6 Replacement of Power Supplies (PS1 & PS2)	33
4.1.7 Replacement of Main Input Circuit Breaker (CB1)	33
4.1.7.1 Replacement of CB1 for units with PN 6550012-07 Rev 02 and After (Manufactured	10/5/17 and Later)
4.1.7.2 Replacement of CB1 for units with PN 6550012-07 Rev A1 and Prior (Manufactured	before 10/5/17)
4.1.8 Replacement of CB1 Handle Mechanism	37

4.1.9 Replacement of CB1 Shunt Trip.....	39
4.1.10 Replacement of Gradient 420V Power Circuit Breaker (CB2).....	43
4.1.11 Replacement of RF-MNS, Pen Cabinet, Supercon Shim PS, and Magnet Power Supply Circuit Breakers (CB3, CB4, CB7, CB8)	45
4.1.12 Replacement of Brainwave, Host, Gradient 48V Control, Recon, and SIO Circuit Breakers (CB5, CB6, CB9-CB11)	45
4.1.13 Replacement of Receptacles and Terminal Blocks (REC1-REC3, TB1-TB3, TB6).....	46
4.1.13.1 Replacement of REC1 and REC2	46
4.1.13.2 Replacement of REC3	47
4.1.13.3 Replacement of TB1	49
4.1.13.4 Replacement of TB2, TB3, and TB6	49
4.1.14 Replacement of Indicators, Switches, and Test Points (IND1-IND3, SW1, SW2, TP1-7)	50
4.1.14.1 Replacement of Indicators IND1-IND3	50
4.1.14.2 Replacement of Switches SW1 and SW2	51
4.1.14.3 Replacement of Test Points TP1-TP7	51
4.1.15 Placement of New FRUs.....	52
4.1.16 Troubleshooting Guide	53
SCHEMATIC DIAGRAM.....	55
APPENDIX A.....	56
REVISION HISTORY	80

IMPORTANT SAFETY INSTRUCTIONS

Even when an EPO/EMO switch is pressed this does not guarantee that the power has been cut off coming into the unit.

To avoid tipping over, do not tip beyond 10° from vertical and avoid abrupt movements.



Voltages in excess of 480VAC are present within this equipment. Turn power off at wall feeder breaker before servicing. Failure to do so could result in serious injury or death.

Des tensions supérieures à 480 V ca sont présents dans cet équipement. Coupez l'alimentation au disjoncteur d'alimentation murales avant de réparer. Ne pas le faire pourrait entraîner des blessures graves ou la mort.



⚠ DANGER



Electrical shock hazard. Lock out power at main panel. Failure to do so could result in serious injury or death.

Risque de choc électrique. Verrouiller alimentation électrique au panneau principal. Ne pas le faire pourrait entraîner des blessures graves ou la mort

⚠ WARNING

Failure to set transformer input taps to match nominal input voltage will result in improper output voltages which may damage equipment, cause load equipment to malfunction or cause injury.

Si vous ne réglez les robinets d'entrée du transformateur pour correspondre à la tension d'entrée nominale se traduira par des tensions de sortie irrégulières qui peuvent endommager l'équipement, l'équipement en charge provoquer un dysfonctionnement ou provoquer des blessures.

⚠ WARNING

Failure to set control transformer to match nominal input voltage may result in equipment damage or malfunction of equipment.

Si vous ne réglez transformateur de commande pour correspondre à la tension d'entrée nominale peut entraîner des dommages matériels ou un dysfonctionnement de l'équipement.

⚠ IMPORTANT

Power conditioner must be properly grounded for safety per National Electric Code, State or Local code.

Conditionneur de puissance doit être correctement mise à la terre pour la sécurité par le National Electric Code, État ou locale de code.

⚠ WARNING

No modification of this equipment is allowed.

Aucune modification de cet équipement est autorisée.

WARNING

To avoid risk of electric shock, this equipment must only be connected to supply mains with protective earth.

Pour éviter tout risque de choc électrique, cet équipement ne doit être raccordé au réseau électrique avec la terre de protection..

WARNING

Connecting electrical equipment to MSOs effectively leads to creating an ME system, and can result in a reduced level of safety.

Connexion d'un équipement électrique pour les OSM conduit effectivement à la création d'un système de ME, et peut résulter en une diminution du niveau de sécurité ..

WARNING

All operators of this unit must have "High Voltage" service training.

Tous les opérateurs de cet appareil doivent avoir une formation "High Voltage" de service.

WARNING

Unit is to be used solely for the purpose in which it was designed.

Unité doit être utilisée uniquement dans le but dans lequel il a été conçu.

Note: To safely terminate operation of the ME equipment, turn off the load, then turn the main breaker off.

Pour en toute sécurité mettre fin au fonctionnement de l'équipement ME, éteignez la charge, puis éteindre le disjoncteur principal.

Note: The supply mains isolation is provided by the CB1 circuit breaker on all poles simultaneously.

L'isolement d'alimentation secteur est fourni par le disjoncteur CB1 simultanément sur tous les pôles. Cette note conforme à la norme AAMI ES60601-1 partie 1 3ème édition.

Note: Assembly of ME systems and modifications during actual service life shall be evaluated based on the requirements of standard AAMI ES60601-1 part 1 3rd edition.

Assemblée des systèmes ME et modifications au cours de la durée de vie réelle doit être évaluée sur la base des exigences de la norme AAMI ES60601-1 partie 1 3ème édition.

Warning Labels:



Line voltage present with machine power off. Risk of electric shock or burn. Turn off & lock out facility power before servicing.

Tension présente ligne avec alimentation de la machine hors tension. Risque de choc électrique ou de brûlure. Éteignez et verrouiller l'alimentation avant l'entretien installation.



Hazardous voltage. Disconnect power before servicing.

Tension dangereuse. Débranchez l'alimentation avant l'entretien.



To reduce the risk of electric shock, do not remove cover. Refer servicing to qualified personnel.

Pour réduire le risque de choc électrique, ne pas enlever le couvercle. Confiez l'entretien à un personnel qualifié.



Class1 ordinary equipment is not suitable for use in the presence of a flammable anesthetic mixture with air or oxygen or nitrous oxide. Rated for continuous operation.

Class1 équipement ordinaire n'est pas adapté pour une utilisation en présence d'un mélange anesthésique inflammable avec l'air ou d'oxygène ou d'oxyde nitreux. Prévu pour fonctionner en continu.



Follow operating instructions. On ME Equipment Follow instructions for use.

Suivez les instructions de fonctionnement. Le ME Equipement Suivez les instructions pour l'utilisation.

 IMPORTANT

As standards, specifications, and designs change from time to time, please ask for confirmation of the information given in this publication.

1- INTRODUCTION

1.1 General Description

The XGD2 Power Distribution Unit (PDU) distributes power to various components of a General Electric Medical Systems MR system while providing surge suppression and electrical isolation for the power line. Three phase input voltage is selectable among 380/400/415/480 VAC at a rated frequency of 50/60 Hz. The power rating for this PDU is 73 kVA continuous and 97 kVA instantaneous. The PDU mounts into the GE 5173027 cabinet.

The design of the MR System is such that there will be no interaction with the PDU during operation by anyone. Not the operator, not the patient and not by anyone else.

This unit can only be operated from the front of the equipment rack that the unit is installed in.

Electromagnetic interference does not affect the operation of this unit.

For questions on the specifics of this unit contact:

TEAL Electronics
10350 Sorrento Valley Road
San Diego, CA 92121
800-888-8325



**XGD2 Power Distribution Unit
Illustration 1-1**

1.2 Scope

This manual describes procedures for installation and basic operation of the XGD2 Power Distribution Unit as well as procedures for failure diagnosis/isolation and installation of field replaceable components of the PDU.

2- SPECIFICATIONS

GE Specifications: 5343114PSP

GE Part Number

5343114

2.1 Electrical Specifications

Power Rating

73 kVA Continuous; 97 kVA Instantaneous

Input Voltage

380/400/415/480 VAC Delta.

Output Voltage

208/120 VAC WYE and 242/420 VAC Wye

Output Circuit Breaker and Connection Requirements

Breaker Size	Function	Ref. Desig	Sec Voltage	Wire to Phase	E-Stop	Customer Access	Connection Type
90A 3 Pole	Gradient 420V Power	CB2	420	ABC	Yes	Front	APP#1321G4, G5(3 Connectors)
63A 3 Pole	RF + MNS	CB3	208	ABC	Yes	Front	Breaker
20A 3 Pole	Pen Cabinet	CB4	208	ABC	Yes	Top Left Side	Breaker
15A 1 Pole	Brainwave	CB5	120	A	No	Top Left Side	Breaker
15A 1 Pole	Host	CB6	120	A	No	Top Left Side	Breaker
32A 3 Pole	Supercon Shim PS	CB7	208	ABC	No	Front	5 wire twist lock (Nema style L21-30R Hubbell) (receptacle front of cab)
32A 3 Pole	Magnet Power Supply	CB8	208	ABC	No	Front	5 wire twist lock (Nema style L21-30R Hubbell) (receptacle front of cab)
10A 2 Pole	Gradient 48V Control	CB9	208	BC	No	Top	7W2 D-Sub connectors
20A 1 Pole	Recon	CB10	120	B	No	Top Rear	NEMA 5-20R receptacle
2A 1 pole	SIO	CB11	120	B	Yes	Top Rear	NEMA 5-20R receptacle

Overload Protection

Input main circuit breaker (CB1). Output distribution circuit breakers.

Load Current

67 Amps total maximum continuous (208V)

68 Amps total maximum continuous (420V), 134A Peak 5 sec Max.

Transformer Impedance

Less than 2%

Efficiency

98% Typical

2.2 Environmental Specifications

Heat Dissipation

4,076 BTU/Hr

2.3 Mechanical Specifications

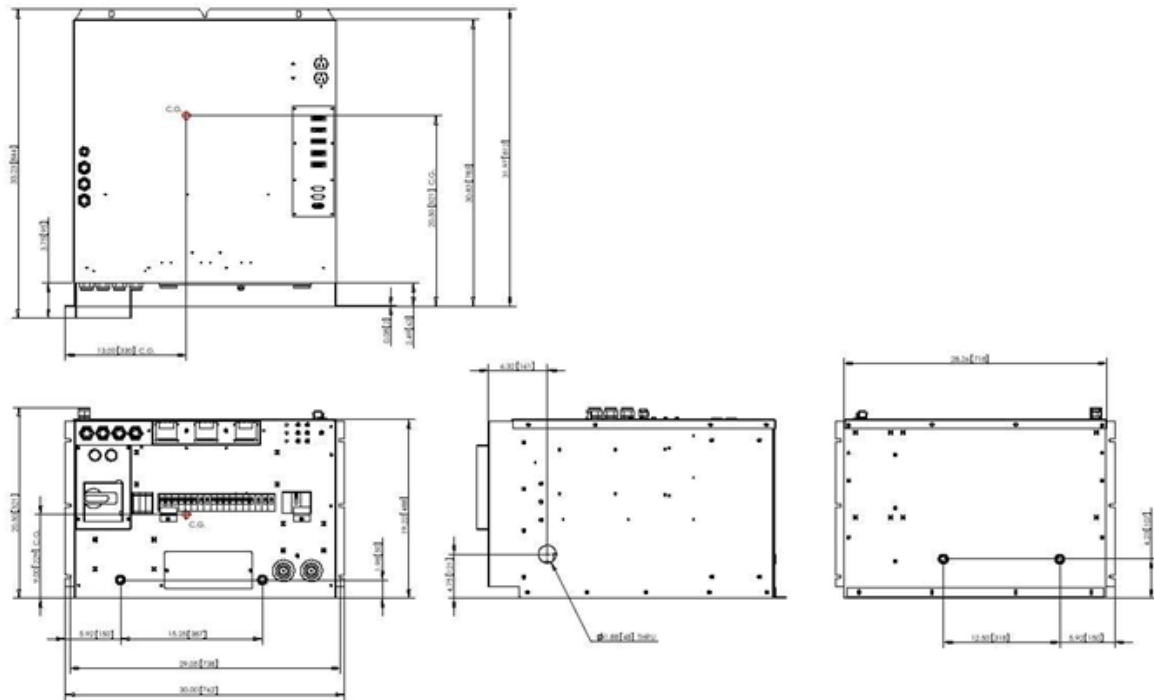
Weight

879 lbs. / 398 kg.

Other environmental Specs

<i>Operating Humidity Range</i>	20-75% non-condensing
<i>Operating Atmospheric Pressure Range</i>	70-106kPa
<i>Transport and storage Temp Range</i>	-20°C to 50°C
<i>Transport and Storage Humidity Range</i>	10-90% non-condensing
<i>Transport and Storage Atmospheric Pressure Range</i>	70-106kPa
<i>Max Temperature Gradient</i>	10°C / hr

Dimensions



**XGD2 Overall Dimensions
Illustration 2-1**

3- INSTALLATION AND OPERATION

3.1 Installation

The installation location must meet the following criteria:

Ambient temp between 0°F and 104°F

Humidity between 0% and 90% non-condensing

Altitude below 6000 feet

Pollution degree 2

Overvoltage Category II

Installation must be performed by a licensed electrician or a factory qualified field engineer.

Unit is intended for use in a controlled environment.

⚠ WARNING

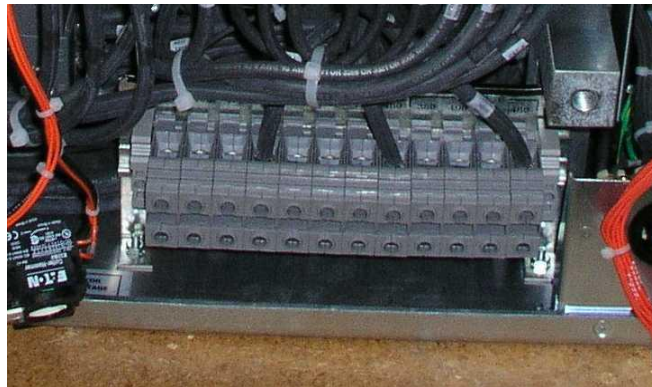
Unit is to be used solely for the purpose in which it was designed.

Unité doit être utilisé uniquement dans le but dans lequel il a été conçu.

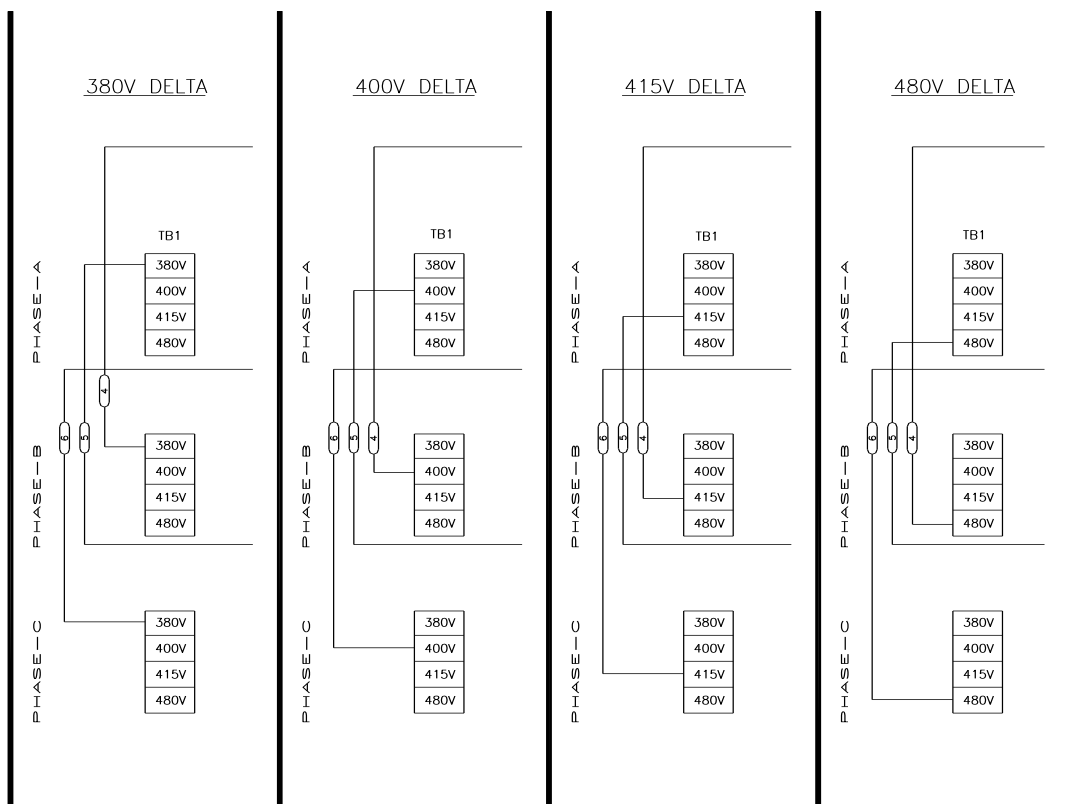
3.1.1 Input Voltage Selection

Before connecting the input power cables, determine the nominal value of the input voltage. The PDU allows for nominal input voltages of 380, 400, 415, or 480 VAC (three phase). Input voltage selection is accomplished as follows:

1. On the Main Disconnect Panel (MDP), turn off the PDU circuit breaker, lock and tag appropriately.
2. Using a voltmeter or other voltage indicating device at the Main Input Terminal Board, check to be sure that no voltage is being applied to the input of the PDU.
3. Remove the small cover panels from the front panel of the PDU marked "INPUT VOLTAGE SELECT".
4. Illustration 3-1 shows the Input Voltage Selection terminal block (TB1). The illustration shows TB1 configured for 480V Delta. For each of the three phases, be sure the wire is connected to the proper terminal for the actual input voltage at the site and that the terminal screw is tight to make a good connection.



INPUT VOLTAGE SELECT TERMINAL BLOCK (TB1)
Illustration 3-1



INPUT VOLTAGE SELECTION
Illustration 3-2

3.1.2 Connection

Note the Cable Access Panel shown in Illustration 3-5 above the main circuit breaker. This panel provides strain relief cable clamps for the input and the various output cables. This panel may be removed from the PDU to provide easier access to the terminal blocks.

1. Route the cables through their respective cable clamps and connect cables to the appropriate terminals as shown.
2. Replace the Cable Access Panel and tighten the cable clamps.



For input cables, use wire sizes in accordance with local regulatory requirements for a 150A load

3.2 Operation

3.2.1 Main Input Circuit Breaker

The Main Input Circuit Breaker operating handle has three positions:

- OFF (as shown in Illustration 3-4),
- ON (vertical), and
- TRIPPED (slightly to the right of vertical). If the Main Input Circuit Breaker has been tripped, it is necessary to move the handle to the OFF position to reset it before turning it ON.



MAIN INPUT CIRCUIT BREAKER OPERATING HANDLE
Illustration 3-4

The Main Input Circuit Breaker may be locked in the OFF position. By pressing on the small triangle on the operating handle, the lockout tab is made accessible to a padlock.

3.2.2 Power Off

This pushbutton at the top of the front panel, when pressed, will immediately trip the Main Input Circuit Breaker, interrupting all power to the load.

3.2.3 EMO Reset

This pushbutton immediately to the right of the Power Off pushbutton will enable the Emergency Stop circuit.

3.2.4 TVSS Indicators

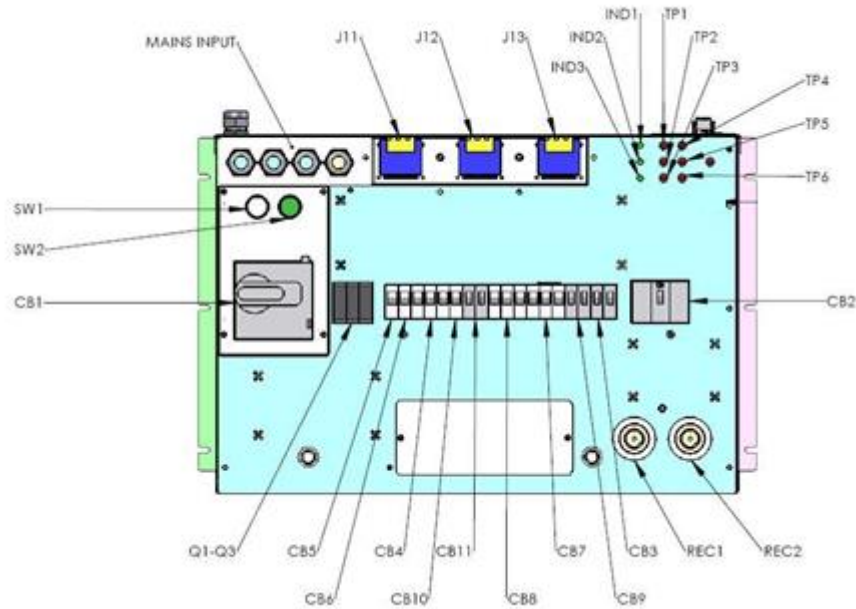
To the right of the Main Input Circuit Breaker are three TVSS (Transient Voltage Spike Suppressor). Any module that fails must be replaced to restore the PDU to operation with surge protection functioning. See Section 4-1.1, Replacement of a TVSS Module, for instructions for replacing a TVSS Module.

3.2.5 Phase Indicators

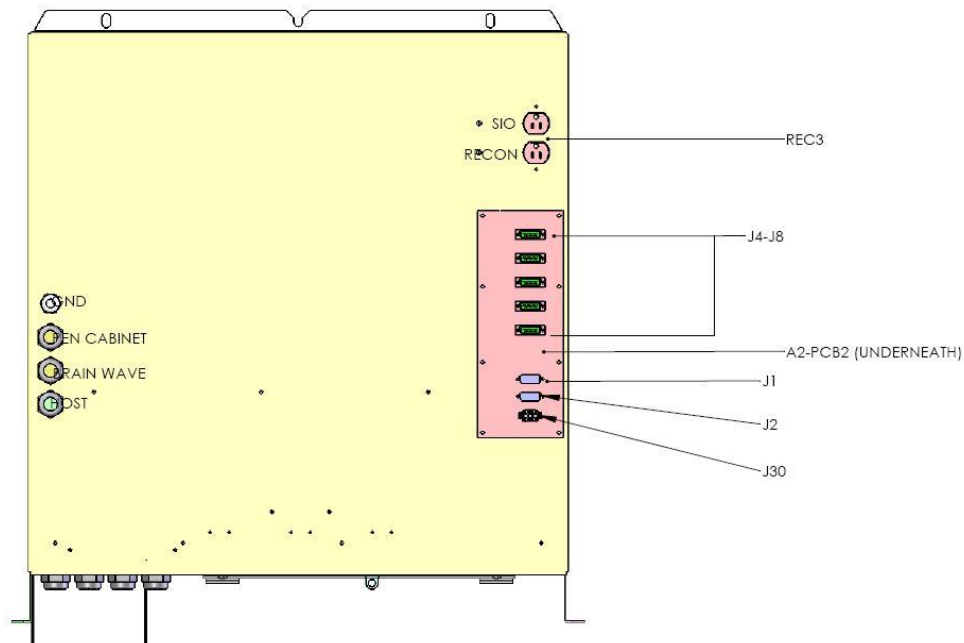
There are three Phase Indicator lights. These indicate that the transformer is energized and ready to deliver power to the load.

3.2.6 Input / Output Voltage Test Points

Near the Phase Indicators on the front panel are jacks into which can be plugged connectors for testing for input and output voltage. See Illustration 3-5. The voltage at these points are reduced by a factor of 100:1 from the actual voltage at the terminal blocks.



FRONT PANEL COMPONENT LOCATIONS
Illustration 3-5



TOP PANEL COMPONENT LOCATIONS
Illustration 3-6

3.3 Closing

3.3.1 Closing Unit Front panel

Anytime the front panel of a unit is removed for **any** reason the following must be verified prior to closing.

- The P6 connector housing is seated **BELOW** the SW1 toggle switch prior to closing
- The SW1 toggle switch on the control board is set in the correct position for the desired configuration as per Table 4.1.3.1 XGD Configuration Settings.

4- FIELD REPLACEABLE UNITS (FRU'S)

No maintenance is required other than periodically (i.e. every 6 months) all electrical connections should be checked for tightness. Accumulations of dust or debris should be removed from front panel to keep ventilation working for convection cooling.

WARNING

All FRU are to be replaced by authorized service personnel only.

Tous les FRU doivent être remplacés par le personnel autorisé seulement.

To replace FRU components – First shutoff and lockout power, remove necessary panels, locate proper component (refer to Location drawing below) and replace with new component (refer to FRU table below). Reinstall all panels and restore power.

The exact type and rating fuse must be used as the replacement, or equipment damage may occur.

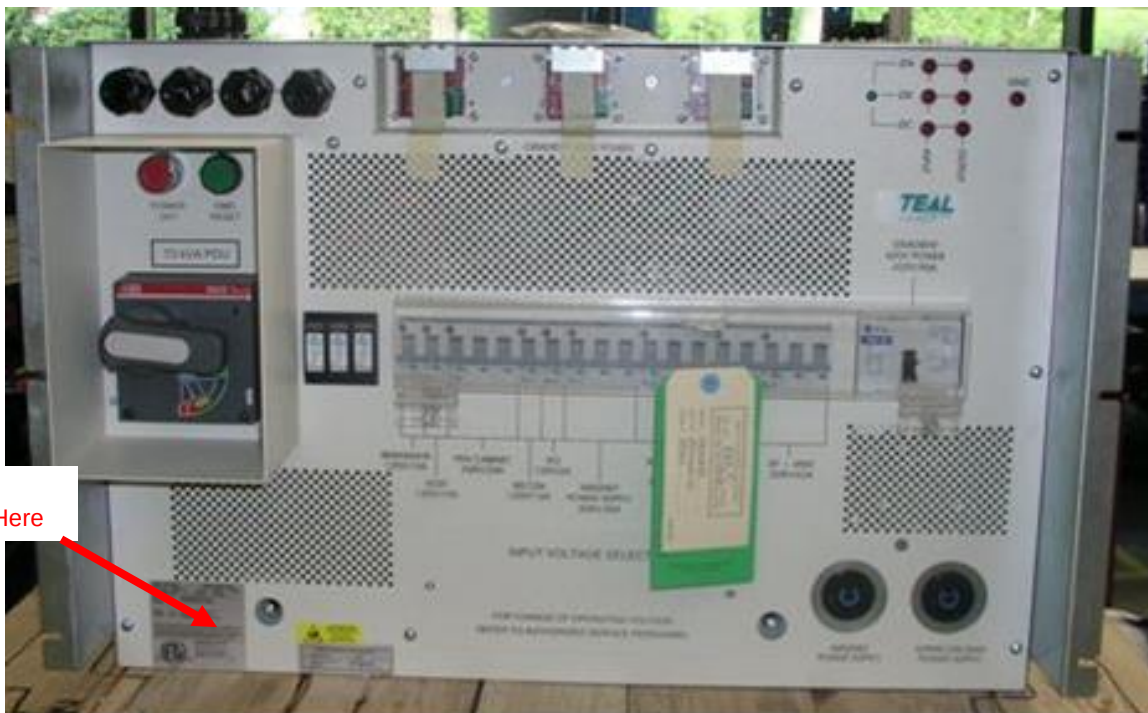
LISTING OF FRU's
4-1

Item	Ref Designator	Description	GE P/N	TEAL P/N
1	A1	Control PC Board AUX Config	5343114-2	4990954
2	A1	XGD PDU2 Control PC Board Asm with PEM or AUX Config Switch	5343114-5	4991072
3	Q1-Q3	TVSS Module FRU for ACGD PDU	2301200-3	4990387
4	CON1	Contactor 65A, 3-Pole	5343114-4	4990950
5	CON2	3 Pole Power Contactor for ESTOP (ABB PN A50-30-11-84, TEAL PN 0410203)	5140621-3	4990781
6	PS1-PS2	48V Power Supplies	5745311	-
7	-	48V Power Supplies Bracket	5343114-10	4991419

8	FAN1-FAN2	Fan With Tach Output Sanyo PN 109E1348G101 (TEAL PN 0860028)	5140621-5	4990783
9	CB1	Main Input Circuit Breaker (EATON - for New PDU Versions)	5789504	4991422
10	CB1	Main Input Circuit Breaker (ABB - for Old PDU Versions)	5604719	4991423
11	-	ABB Shunt Trip (for old PDU Versions)	5823570	4991525
12	-	ABB Handle Mechanism (for old PDU Versions)	5823571	4991526
13	-	EATON Shunt Trip (for new PDU Versions)	5823568	4991528
14	-	EATON Handle Mechanism (for new PDU Versions)	5823569	4991529
15	CB2	Gradient 420V Power Circuit Breaker	5789505	4991424
16	CB3-4, CB7-8	RF-MNS, Pen Cabinet, Supercon Shim PS, and Magnet Power Supply Circuit Breakers	5789506	4991426
17	CB5-6, CB9-11	Brainwave, Host, Gradient 48V Control, Recon, and SIO Circuit Breakers	5789507	4991427
18	REC1-3, TB1-3, TB6	Receptacles and Terminal Blocks	5789508	4991437
19	IND1-3, SW1-2, TP1-7	Indicators, Switches, and Test Points	5789509	4991438

4.1 FRU Installation

Before moving forward with FRU replacement, you will need to identify the unit part number (PN) and revision of the unit being worked on as some FRU instructions are dependent on the unit requiring replacements. Proceed to the appropriate section that corresponds with the unit if a FRU has multiple instructions.



Find Unit PN
and Revision Here

Versions of the PDU changed by incrementing the part number (i.e. 6550012-05 to 6550012-06 to 6550012-07, and so on) and by Revision. Older PDUs used alpha numeric revision numbers (A0-A9, B0-B9, etc.) while revisions made during part of 2016 and newer use numeric revision numbers (00, 01, 02, etc.). In some cases, PNs may contain both alpha numeric and numeric revision numbers (i.e. A0, A1, A2, 03, 04, and so on).

4.1.1 Replacement of the TVSS Module

The operation of the TVSS Indicators is explained in Section 3-2.4, TVSS Indicators. If one (or more) of the indicators shows that a TVSS Module has failed, follow the steps below to replace the failed module and restore TVSS protection:

1. Turn off the PDU circuit breaker, lock and tag.
2. Using a voltmeter or other voltage indicating device, check the voltage at the Main Input Terminal Board to be sure that no voltage is being applied to the input of the PDU.
3. Pull out the failed TVSS Module(s) and plug in the replacement(s) as shown in Illustration 4-1.



TVSS MODULE REPLACEMENT
Illustration 4-1

4. Re-energize the PDU.
5. Verify that all TVSS Modules are good (no red flags displayed).

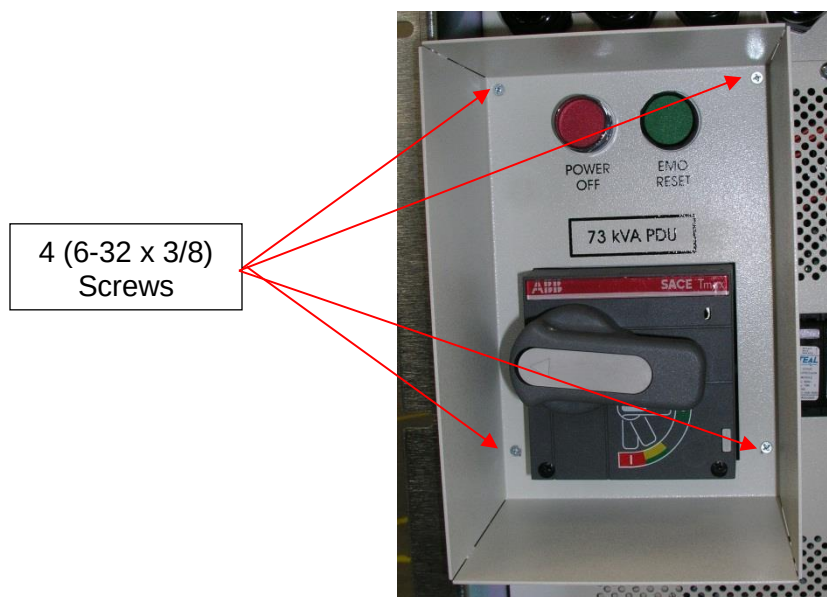
4.1.2 Removal of Front Panel to Access Additional FRUs

Tools needed: Phillips screwdriver or power screwdriver set, 11/32 open wrench.

1. Turn off the main input circuit breaker by rotating the lever to the off position. Perform the lockout-tagout (LOTO) procedure at the MDP.

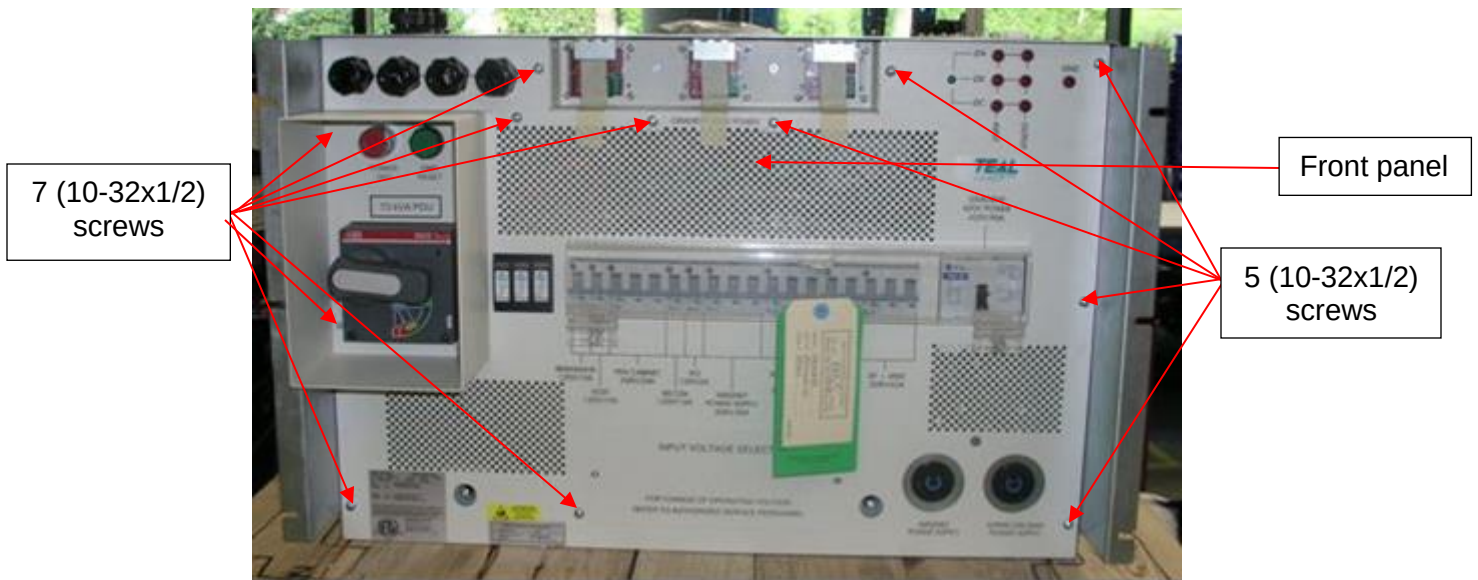


2. Remove the 4 (6-32 x 3/8) screws on the circuit breaker guard using a Phillips screwdriver.
Note: Do not discard screws, as they will be used to reattach the guard back over the circuit breaker.



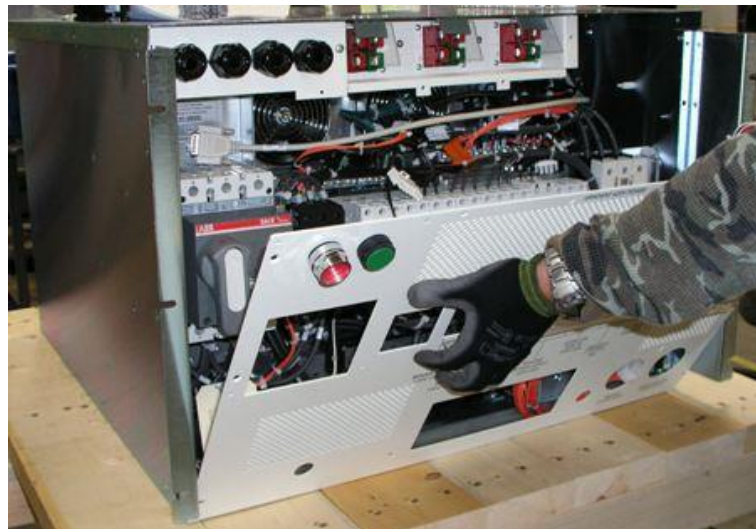
3. Remove the 12 (10-32x1/2) screws on the front panel using Phillips screwdriver.

Note: Do not discard screws, as they will be used to reattach the front plate.



4. Slightly pull the front panel and lean it down and away from the unit.

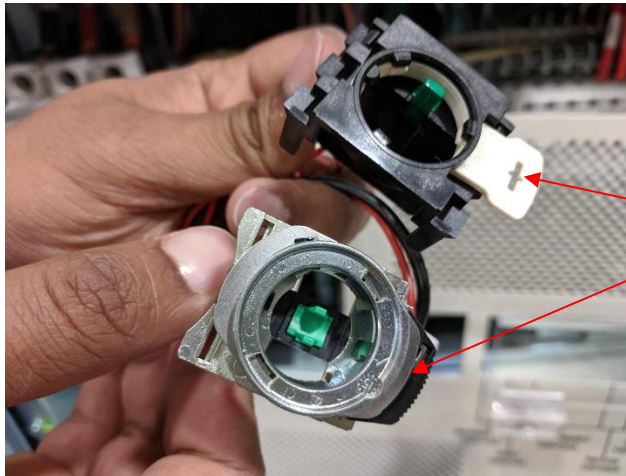
Note: Be careful not to damage the leak sensor strip.



5. Disconnect SW1 and SW2 from the back of the front panel.

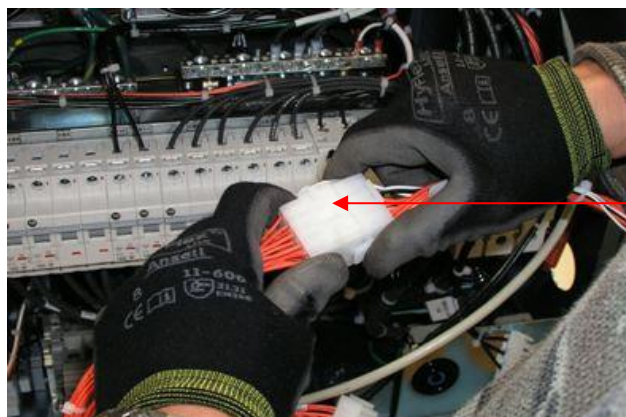
Note: No tools are required. Turn tabs counterclockwise and gently pull assembly apart.

SW1 and SW2



Tabs

6. Disconnect the P8 and J8 15-pin housing connectors.
Note: No tools required. Press down on the locking tabs and pull connectors apart.



15-pin housing
connectors (P8/J8)

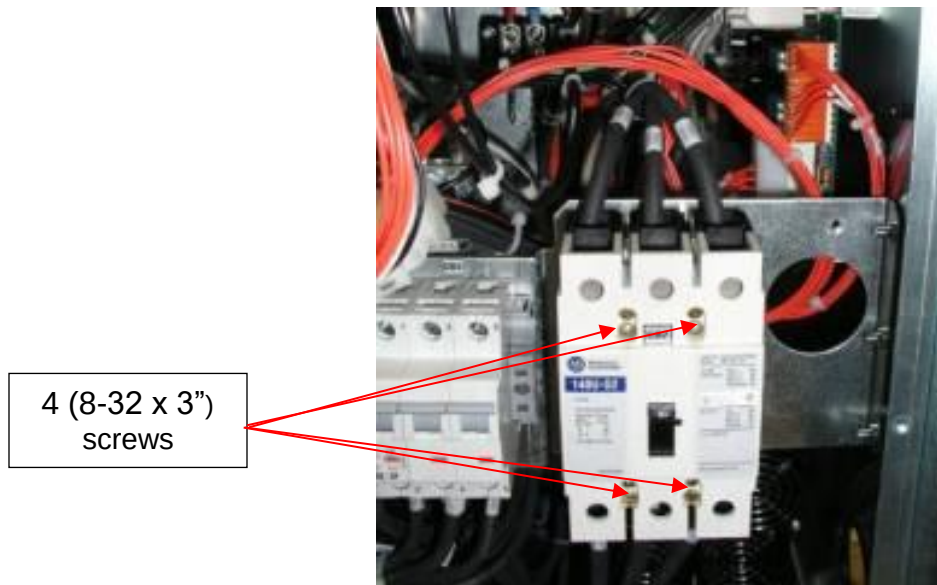
4.1.3 Replacement of the Control PC Board

If the control board is deemed defective as per the Troubleshooting Guide in 4.1.16 follow the steps below to replace the control board. The control board is mounted on the inside of the side panel.

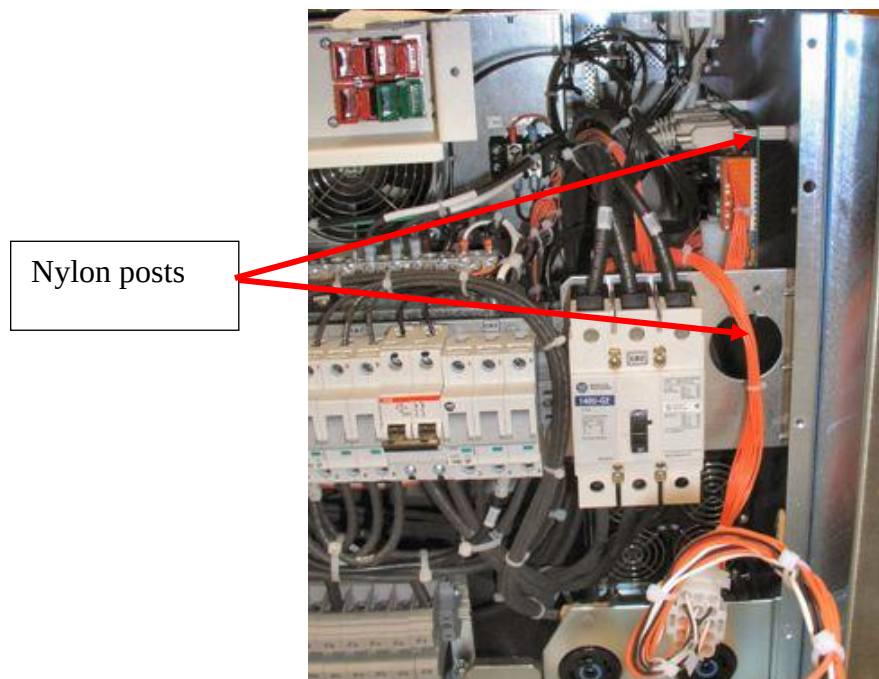
Circuit boards or assemblies with static sensitive devices will be packaged in accordance to prevent damage due to ESD (Electrostatic Discharge). Procedures to prevent ESD should be followed when handling static sensitive devices

1. Remove the front panel as per 4.1.2
2. Remove the four mounting screws from the circuit breaker (CB2) and move the CB to the side.

Note: Do not discard screws, as they will be used to replace CB.



3. Using the tool supplied with the PCB, pop the PCBA off the front two nylon posts. the lower post can be accessed through the port hole



4. With an ESD strap on, gently slide the PCBA out of the rear holders. Ensure that the toggle position of SW1 is properly set and has not been altered during the removal of the board.

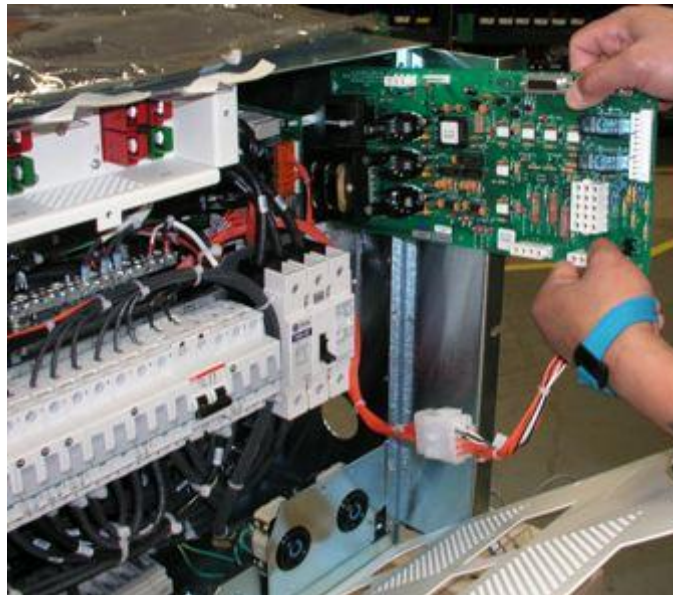


Figure 4.1.3.4

5. Disconnect cables P1, P2, P3, A1-P5, A1-P6, A1-P7, A1-P8. Mark the cables with a masking tape and marker to ensure correct position when re-connecting the cables.
6. Configure the replacement of the PDU Control board according to the cabinet type. All boards should be shipped with the board set to the AUX position. Replace the board according to the instructions in 4.1.7 (the above instructions in reverse order).
7. When reinstalling the board ensure that P6 connector bundle is seated behind porthole and beneath the PC Control board toggle switch.

Table 4.1.3.1 XGD Configuration Settings

Board Part #	Jumper / Switch	LED Color	Position
5343114-2	JP1 and JP2	N/A	1-2
5343114-5	SW1	Green	Switch “UP” when viewed from front. LED3 is green.

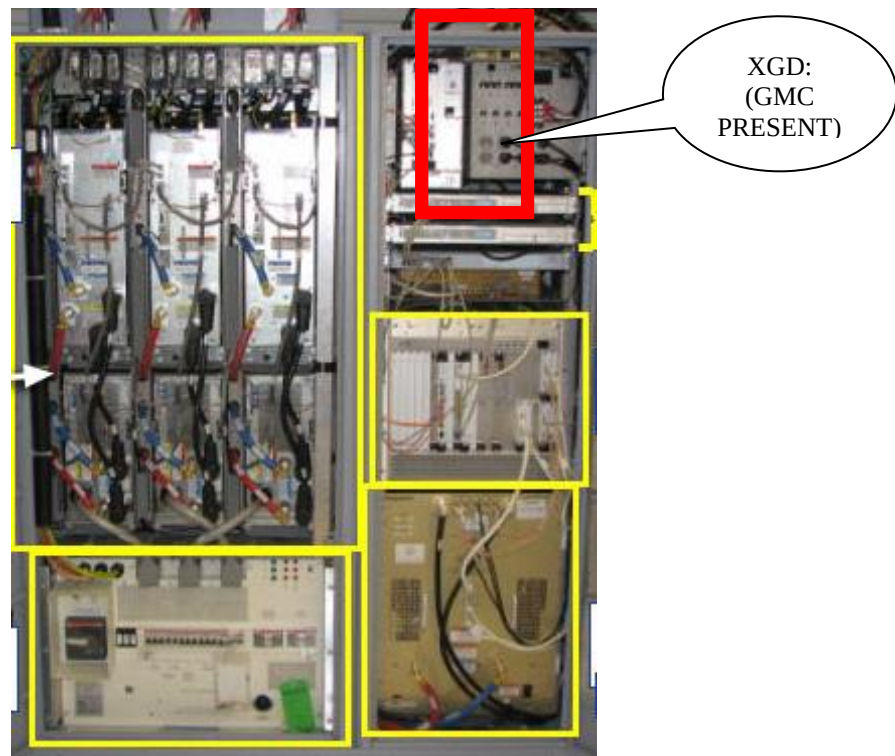


Figure 4.1.3.1 XGD Cabinet

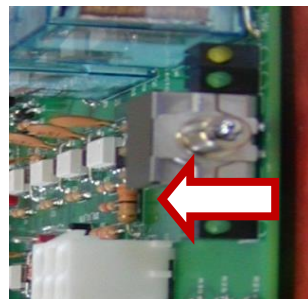
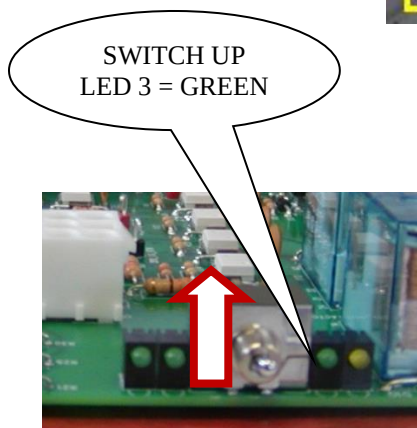


Figure 4.1.3.2 XGD Configuration
5343114-5 Control Board
(No Jumpers)

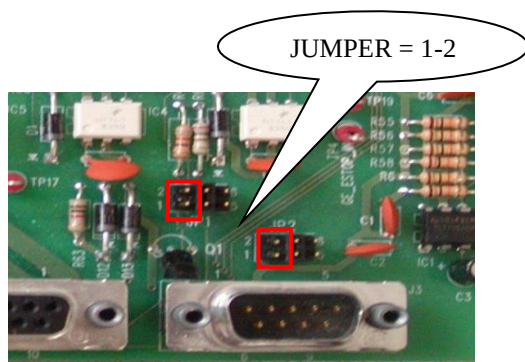


Figure 4.1.3.3 XGD Configuration
5343114-2 Control Board
(No Switch)

Table 4.1.3.2 XGD II Configuration Settings

Board Part #	Jumper / Switch	LED Color	Position
5343114-2	JP1 and JP2	N/A	3-4
5343114-5	SW1	Yellow	Switch "DOWN" when viewed from front. LED4 is yellow.

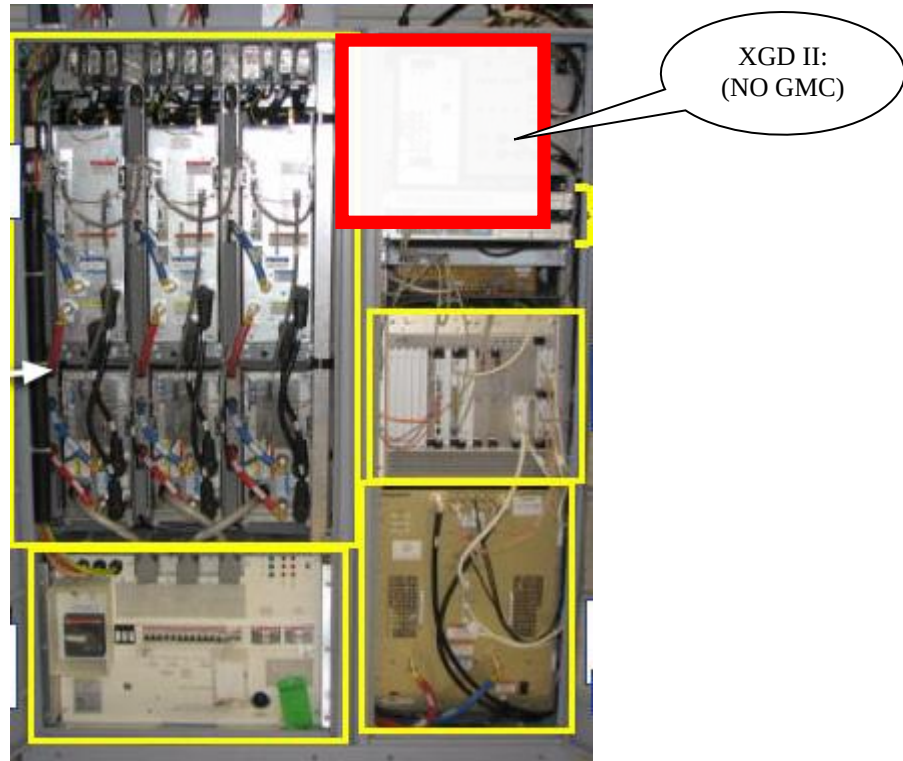
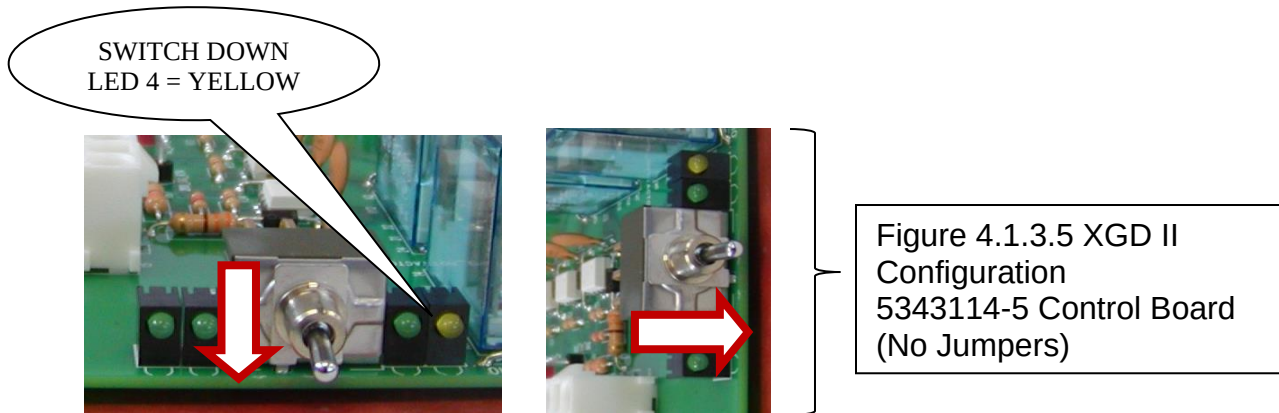


Figure 4.1.3.4 XGD II Cabinet



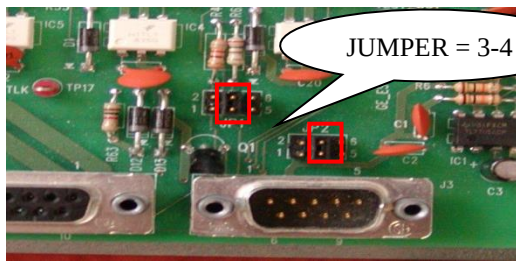
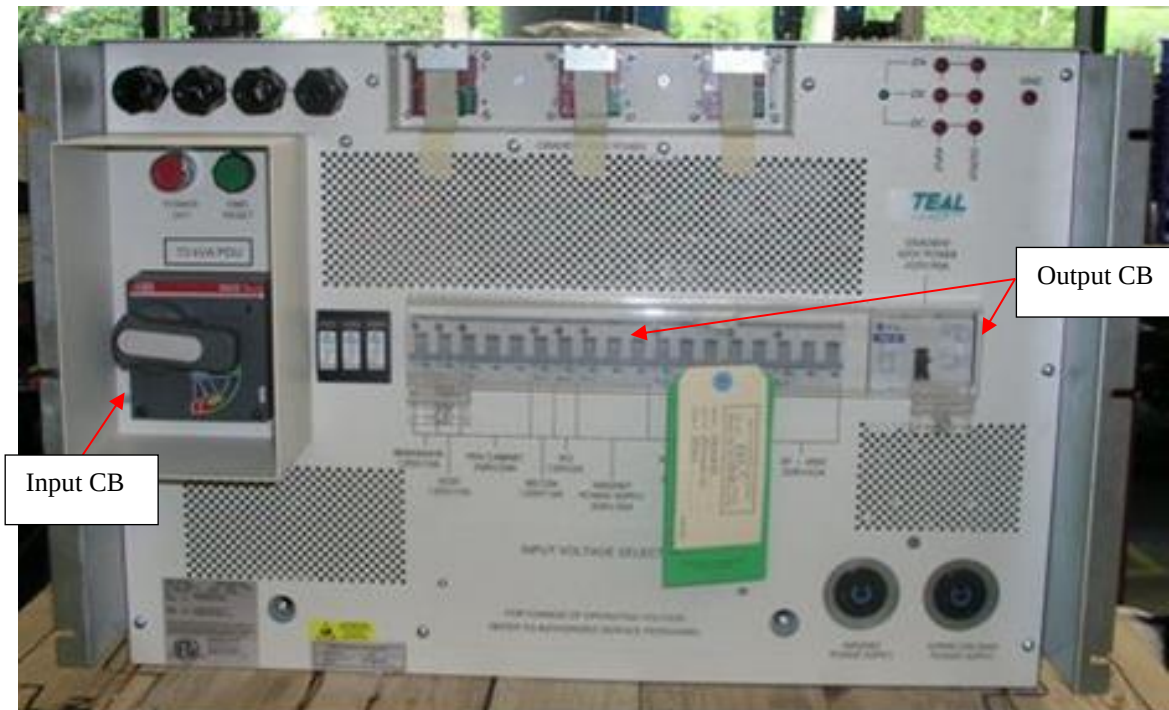


Figure 4.1.3.6 XGD II Configuration
5343114-2 Control Board
(No Switch)

8. The next three steps will verify you have properly replaced the PDU control board.
TURN OFF ALL CIRCUIT BREAKERS ON THE PDU BEFORE CONTINUING TO AVOID UNNECESSARY POWER CYCLING TO OTHER SYSTEM COMPONENTS.



9. Once the front cover is re-installed, apply power to PDU and PGR cabinet by turning on the main input circuit breaker (CB1) and then the Gradient 48V Control circuit breaker (CB9) to power up the PEM/AUX Module. Once powered is applied, you may inspect the control board through the upper right vent holes in the front cover and verify the appropriate configuration LED is on. This LED is located above the configuration switch.
- Green indicates XGD configuration
 - Yellow indicates XGD II configuration.

NOTICE

Incorrect replacement or configuration of the PDU Control Board may immediately activate Emergency Off and cause main PDU CB1 to trip and open.

This will happen if the PDU Control Board with PEM configuration is installed inside an XGD PGR cabinet which contains an XGD Auxiliary “AUX” Module. In this situation, when the main PDU CB1 is closed, incoming power is applied the PDU Control Board. Once this board boots, it will immediately activate Emergency Off and cause main PDU CB1 to trip and open.

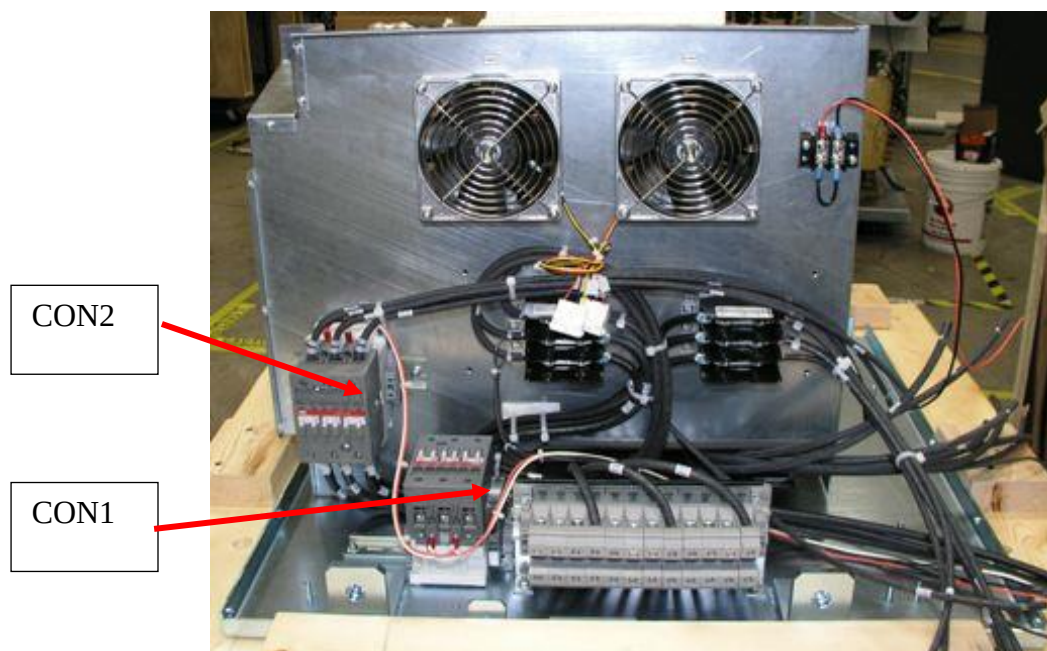
10. After configuration confirmation, perform the leak sensor functionality check in accordance with the “System Power On” service direction under the installation -> system folder located within the product’s service methods. Re-check PDU Control board set-up and cabling if this fails.
11. Verify the functionality of the E-stop circuitry in accordance with the “Emergency Stop and Off Check” service direction under the Functional Checks -> System folder located within the product’s service methods. Once these tests are completed, you may apply power to the rest of the components in the system.

4.1.4 Replacement of Contactors (CON1 or CON2)

If one or more of the contactors are deemed defective as per the Troubleshooting Guide in 4.1.16 follow the steps below to replace the contactors. The contactors are mounted behind the circuit breaker panel of the PDU located on the front panel.

Contactors or assemblies with static sensitive devices will be packaged in accordance to prevent damage due to ESD (Electrostatic Discharge). Procedures to prevent ESD should be followed when handling static sensitive devices.

Tools needed: Phillips screwdriver or power screwdriver set, slotted screwdriver, 11/32 open/closed or socket wrench.



1. Remove the front panel as per 4.1.2

2. Disconnect wires 97, 98 at the AUX of CON1 and wires 200 – 202 at T1-T3 of CON1. Make sure you mark the wires with a masking tape and marker.
3. Snap out CON1 from the mounting bracket and disconnect wire 148 – 150 at L1 – L3.
Note: *To snap out ABB contactors, pull the tab at the bottom of the contactor to release it from the din rail. To snap out Eaton contactors, simply pull it off the din rail.*
4. Disconnect wires 69/180 at A1 of CON1 and wire 147/181 at A2 of CON1. Make sure you mark the wires with a masking tape and marker.
5. Remove CON1 from unit and proceed to removing CON2.
6. Disconnect wire 151 at L1, 152 at L2 and 153 at L3 of CON2.
7. Snap out CON2 and disconnect wires 128/63 at T1, 65/129 at T2 and 130/66 at T3 of CON2.
Note: *To snap out ABB contactors, pull the tab at the bottom of the contactor to release it from the din rail. To snap out Eaton contactors, simply pull it off the din rail.*
8. Disconnect wire 180 at A1 and 181 at A2 of CON2.
9. Remove CON2 from unit and prepare to install the replacement parts. Replace old contactor with a new contactor. Re-connect the wires on CON1 and or CON2 based on schematic diagram. Tighten all connections. Torque specification for T1-T3, L1-L3 is 35 in. lbs. And torque specification for AUX and A1-A2 is 9 in. lbs.

4.1.5 Replacement of Fans (FAN1 or FAN2)

Before moving forward with circuit breaker replacement, you will need to identify the unit part number (PN) and revision of the unit being worked on. Proceed to the appropriate section below that corresponds with the unit.



Find Unit PN
and Revision Here

If the fans are deemed defective as per the Troubleshooting Guide in 4.1.16 follow the steps below to replace the fans. The fans are mounted behind the circuit breaker panel of the PDU located on top of the unit.

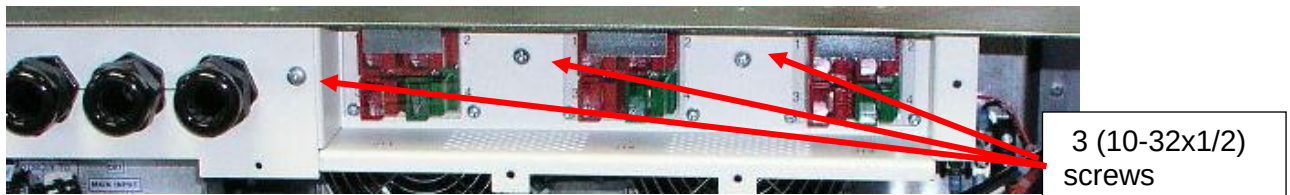
Fans or assemblies with static sensitive devices will be packaged in accordance to prevent damage due to ESD (Electrostatic Discharge). Procedures to prevent ESD should be followed when handling static sensitive devices.

4.1.5.1 Replacement of Fans (FAN1 or FAN2) for units PN 6550012-06 Rev A3 and Prior

Tools needed: Phillips screwdriver or power screwdriver set
5/16 open end wrench (small)

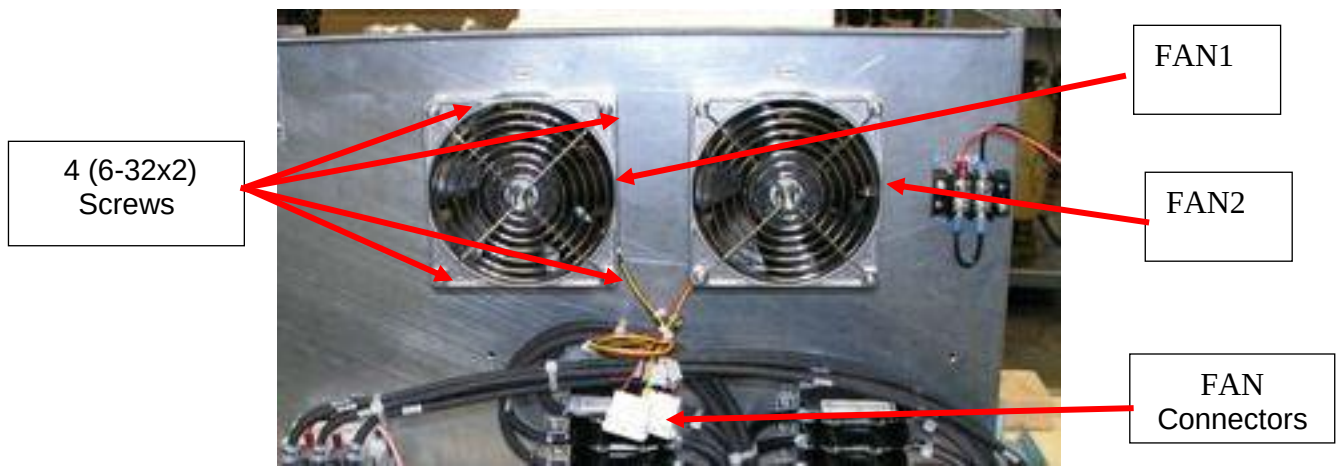
CAUTION - Care must be taken so that hardware does not fall down into the unit.

1. Remove the front panel as per 4.1.2
2. Remove 3 (10-32x1/2) screws at the Anderson assembly connector. Note: Do not discard screws, they will be used to replace assembly.

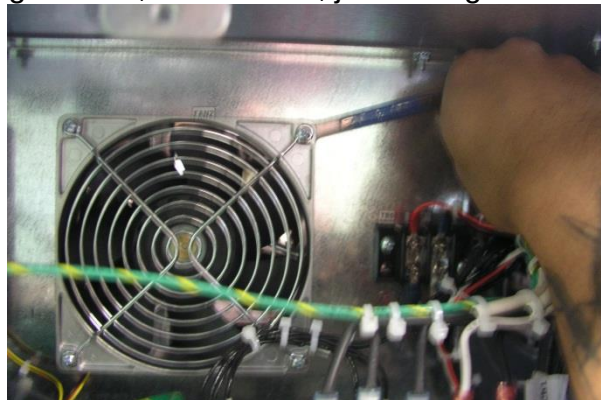


3 (10-32x1/2)
screws

3. Remove fans (FAN1 and FAN2) by unscrewing 4 (6-32 x 2) screws using Phillips screwdriver. Note: Do not discard screws; they will be used to replace fan assembly.



4. Unplug fan connectors.
5. If left fan (FAN1) needs replacing, right fan (FAN2) must be removed first to allow wrench access. Using wrench, loosen nuts, just enough to break the lock, on the fan.

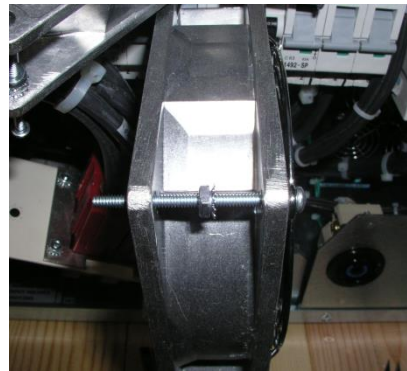
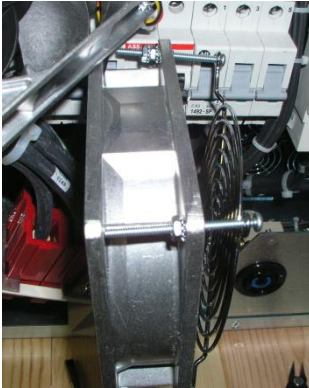


6. Use Philips head screw driver to back out the screws, just to the point that the fan separates from the back panel.
7. Once the fan on the right is completely removed you can now access the nuts on the left fan, following the instructions above.



8. When the old fans are out and away from the unit, remove the screws completely retaining ONE of the locking nuts and the fan guard.

9. Place the fan guard on the front of the new fan; slide the screw through the front hole. Place the locking nut so it will sit about half way once the screw is through the back hole.



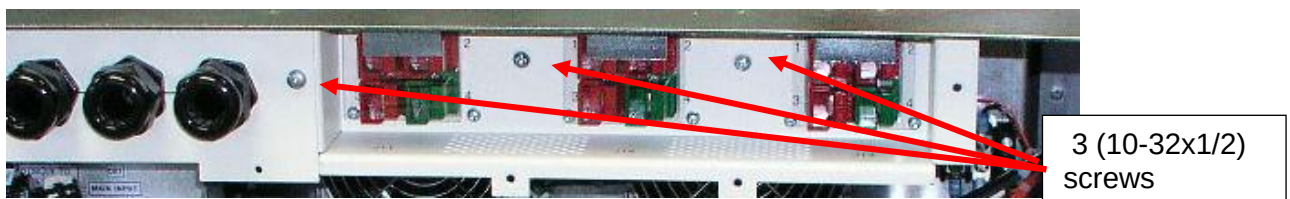
10. There is no need to tighten the nut. It is used to retain the screw and the guard upon removing or placement of the fan to prevent hardware from falling down into the electronics below.
11. Place new fan assembly and screw on to plate. Re-connect housing connectors P4 with J4 and P5 with J5.
12. Replace panels removed using screws that were set aside.

4.1.5.2 Replacement of Fans (FAN1 or FAN2) for units PN 6550012-06 Rev A4 and After

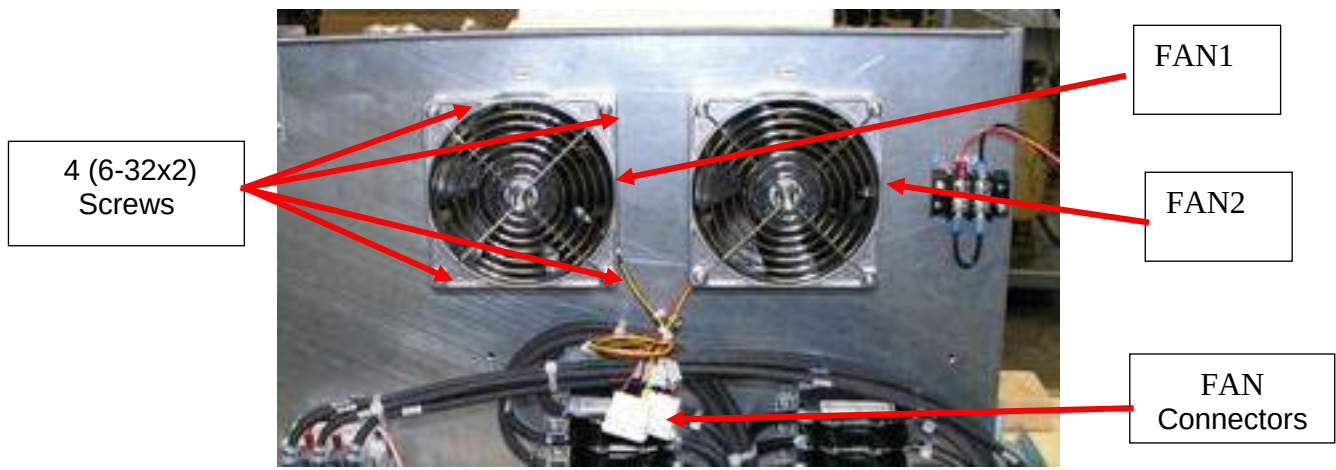
Tools needed: Phillips screwdriver or power screwdriver set

CAUTION - Care must be taken so that hardware does not fall down into the unit.

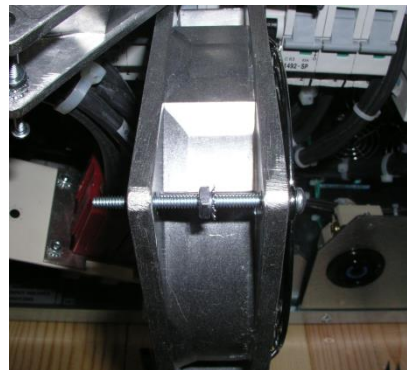
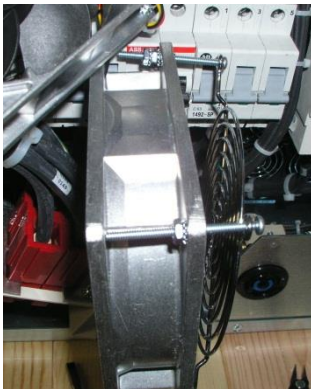
1. Remove the front panel as per 4.1.2
2. Remove 3 (10-32x1/2) screws at the Anderson assembly connector. Note: Do not discard screws; they will be used to replace assembly.



3. Remove fans (FAN1 and FAN2) by unscrewing 4 (6-32 x 2) screws using Phillips screwdriver. Note: Do not discard screws; they will be used to replace fan assembly.



4. Unplug fan connectors.
5. Use Philips head screw driver to back out the screws, just to the point that the fan separates from the back panel.
6. When the old fans are out and away from the unit, remove the screws completely retaining the locking nut and the fan guard.
7. Place the fan guard on the front of the new fan; slide the screw through the front hole. Place the locking nut so it will sit about half way once the screw is through the back hole.



8. There is no need to tighten the nut. It is used to retain the screw and the guard upon removing or placement of the fan to prevent hardware from falling down into the electronics below.
9. Place new fan assembly and screw on to plate. Re-connect housing connectors P4 with J4 and P5 with J5.
10. Replace panels removed using screws that were set aside.

4.1.6 Replacement of Power Supplies (PS1 & PS2)

If the power supplies are deemed defective as per the Troubleshooting Guide in 4.1.16 follow the steps outlined in Appendix A. The power supplies are mounted behind the receptacle assembly of the PDU located on bottom of the unit.

4.1.7 Replacement of Main Input Circuit Breaker (CB1)

Before moving forward with circuit breaker replacement, you will need to identify the unit part number (PN) and revision of the unit being worked on. Proceed to the appropriate section below that corresponds with the unit.

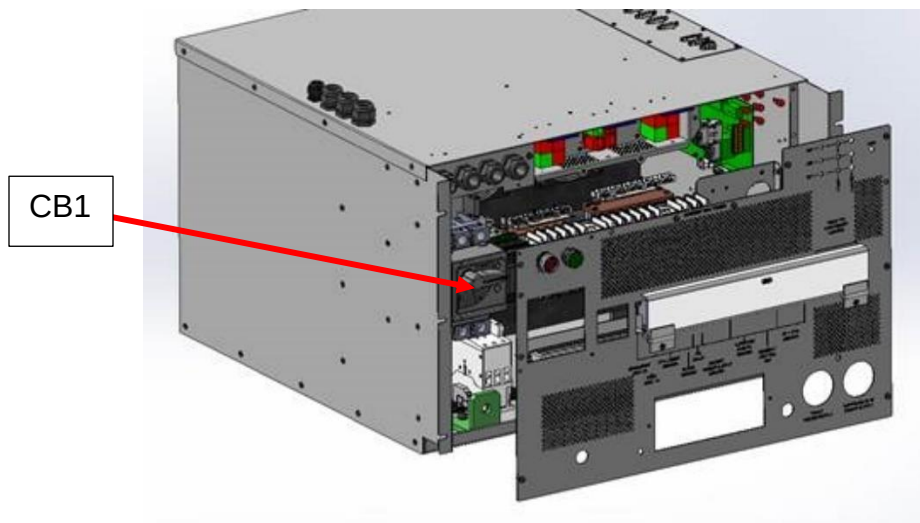


If the circuit breaker is deemed defective as per the Troubleshooting Guide in 4.1.16 follow the steps below to replace the circuit breaker.

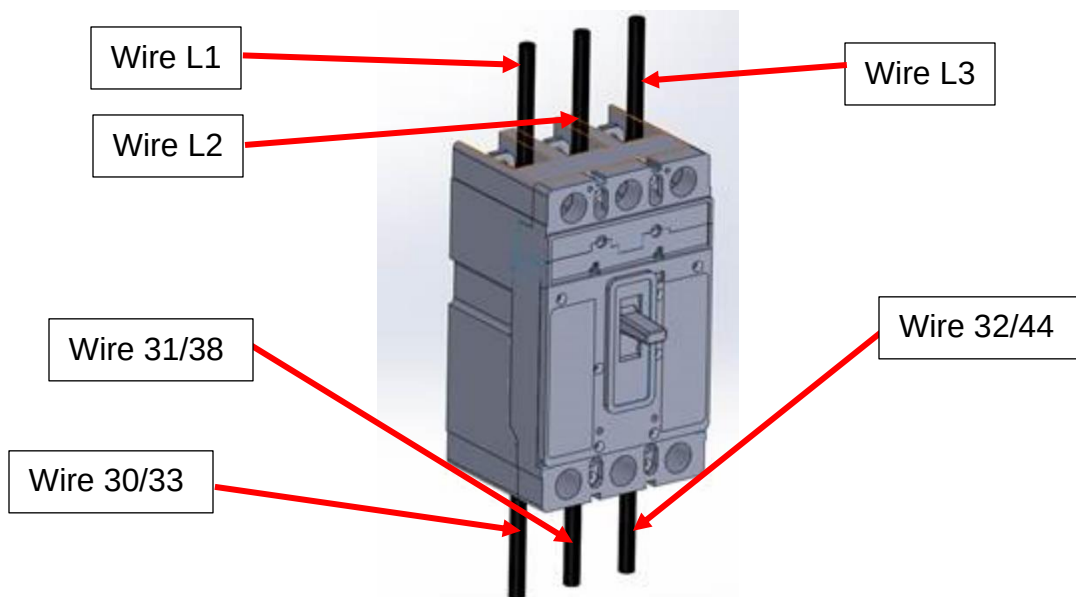
4.1.7.1 Replacement of CB1 for units with PN 6550012-07 Rev 02 and After (Manufactured 10/5/17 and Later)

Tools Needed: Phillips screwdriver, hex key, open wrench

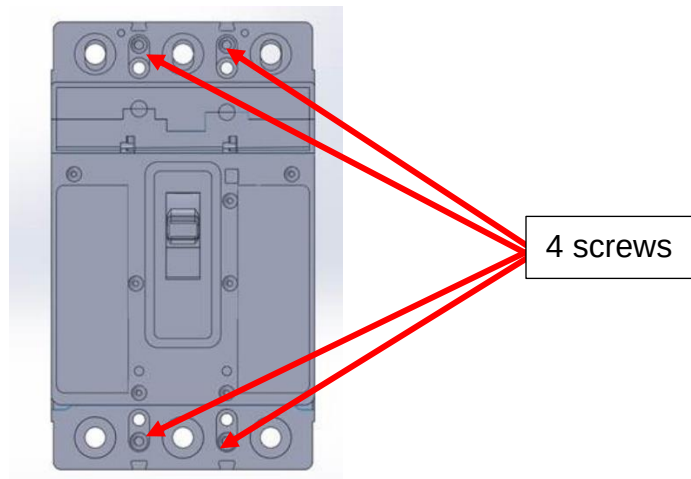
1. Remove the front panel per section 4.1.2.



2. Disconnect wires 30/33, 31/38, and 32/44 from the output (bottom) of the Eaton circuit breaker CB1 using a hex key. Disconnect wires L1, L2, and L3 from the input (top) of the breaker. Label wires with masking tape if necessary.



3. Cut shunt trip wires 70 and 73.
4. Remove the 4 mounting screws holding the circuit breaker to the plate.

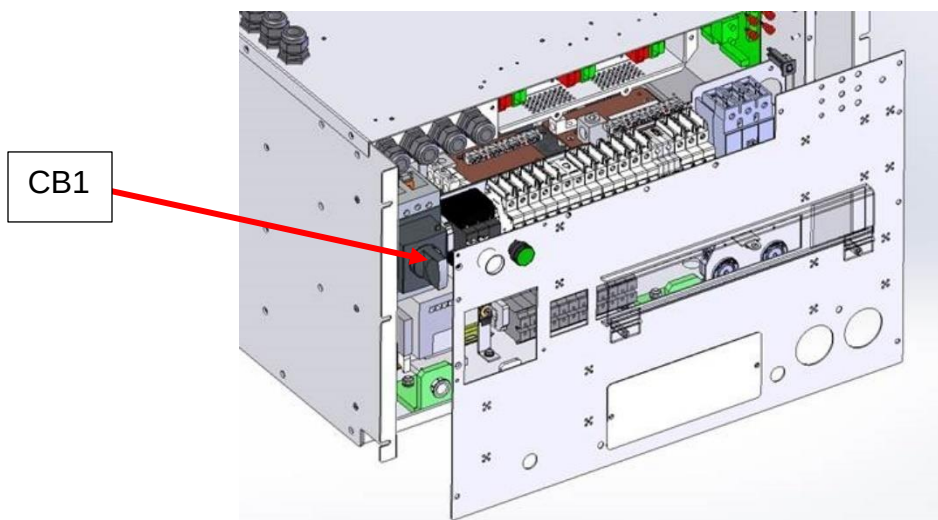


5. Remove the circuit breaker from the mounting plate.
6. Install the new Eaton circuit breaker in place (4 screws).
7. Butt splice the new shunt trip wires.
8. Reconnect the wires removed in step 2.

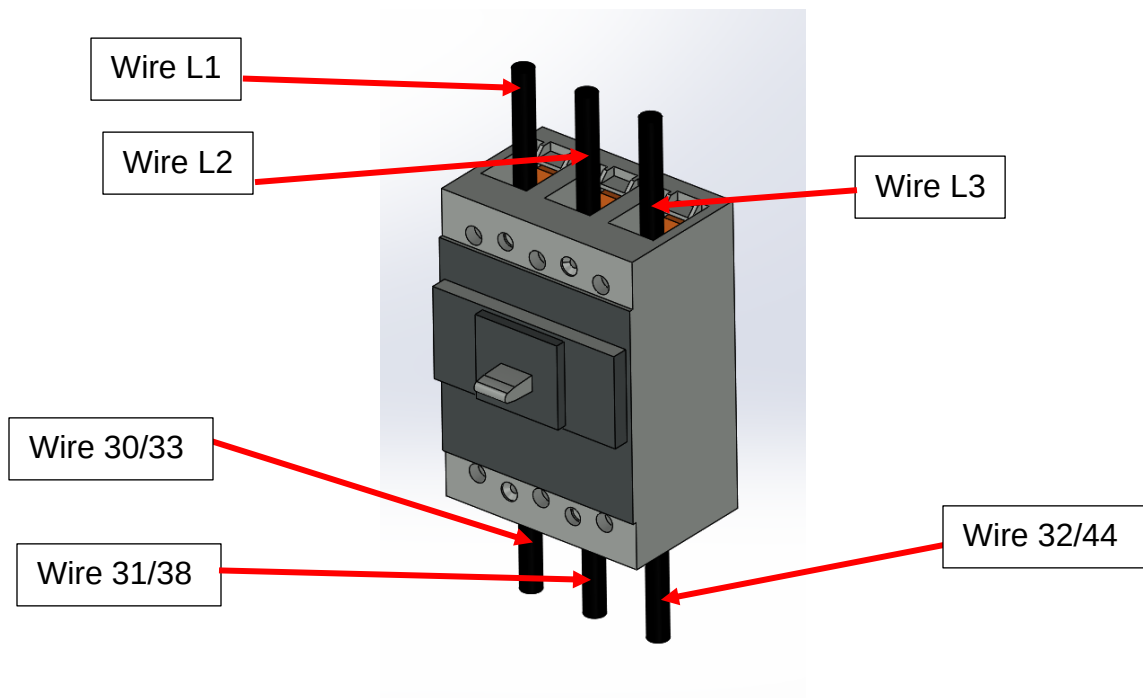
4.1.7.2 Replacement of CB1 for units with PN 6550012-07 Rev A1 and Prior (Manufactured before 10/5/17)

Tools Needed: Phillips screwdriver, open wrench

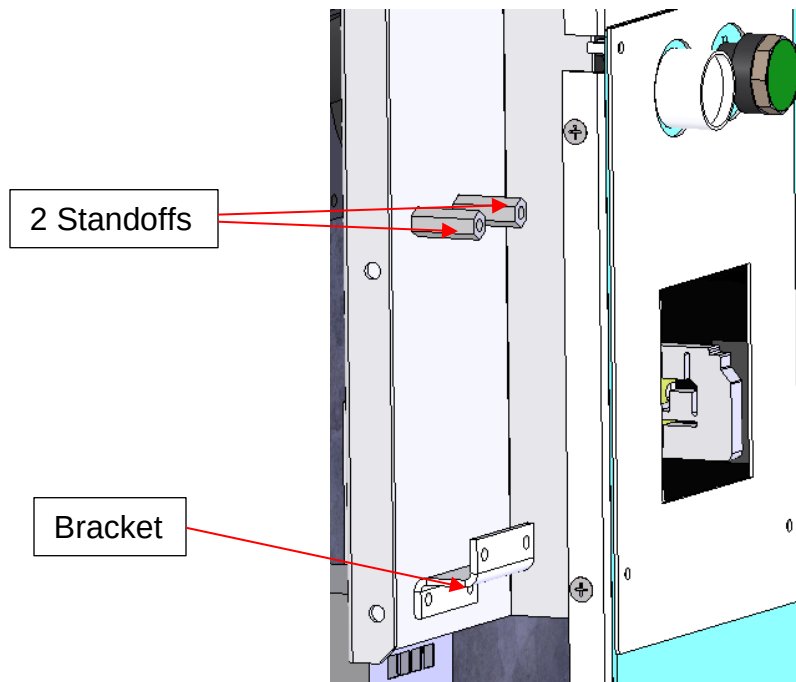
1. Remove the front panel per section 4.1.2.



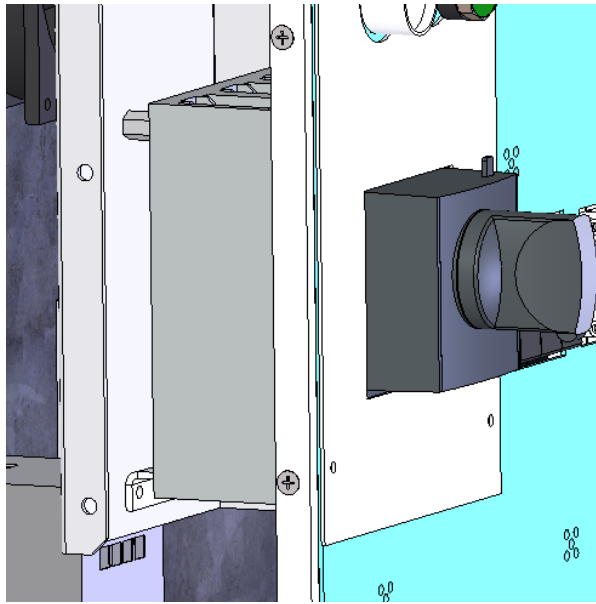
2. Disconnect wires 30/33, 31/38, and 32/44 from the output (bottom) of the ABB circuit breaker CB1 using a hex key. Disconnect wires L1, L2, and L3 from the input (top) of the breaker. Label wires with masking tape if necessary.



3. Cut shunt trip wires 70 and 73.
4. Remove the 4 mounting screws holding the circuit breaker to the plate.
5. Remove the circuit breaker from the mounting plate.
6. Install the provided standoffs and bracket to hold the new CB in place.



7. Install the new ABB circuit breaker in place (4 screws).

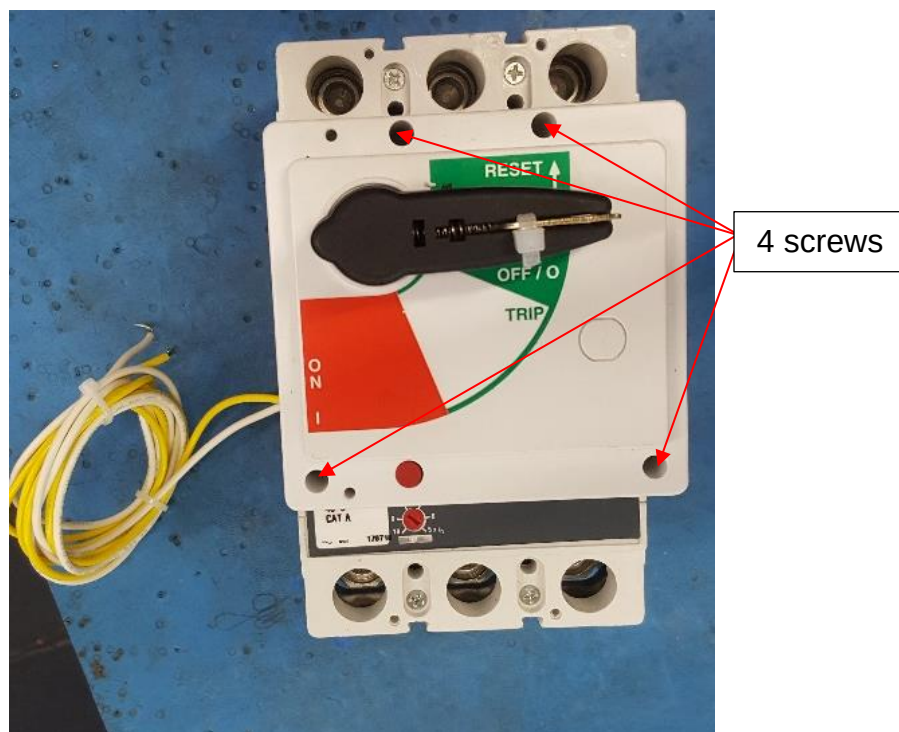


8. Butt splice the new shunt trip wires.
9. Reconnect the wires removed in step 2

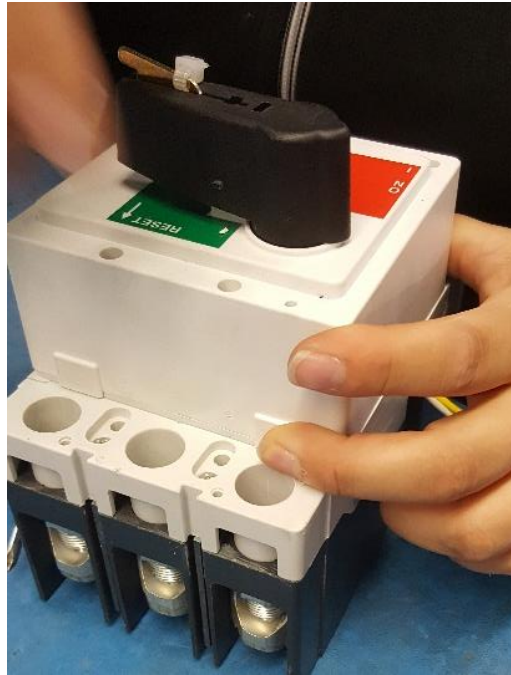
4.1.8 Replacement of CB1 Handle Mechanism

If the circuit breaker handle mechanism is deemed defective as per the Troubleshooting Guide in 4.1.16 follow the steps below to replace the circuit breaker.

1. On the front of CB1, remove the 4 screws holding the circuit breaker handle in place.



2. Remove the handle from the circuit breaker.



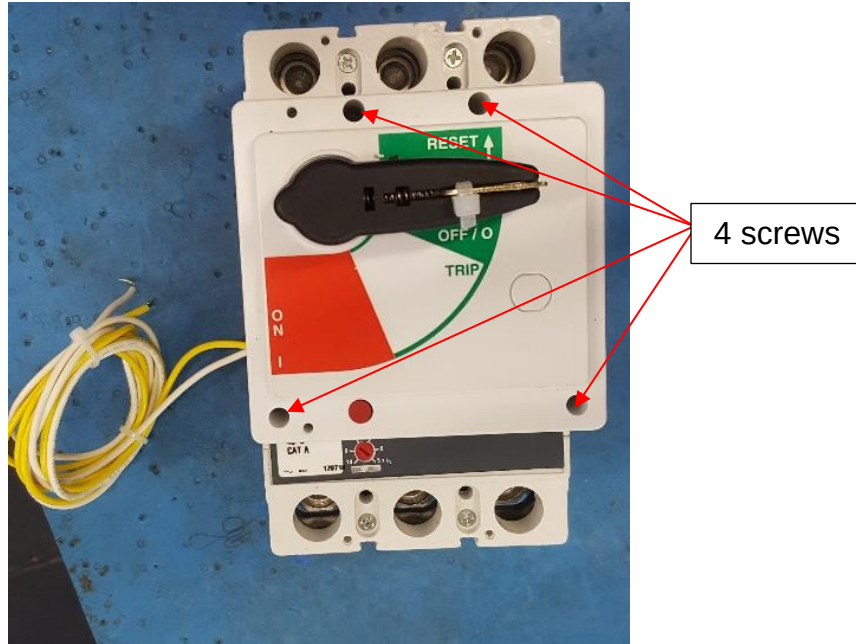
3. Reinstall the new handle with the screws removed in step 1.



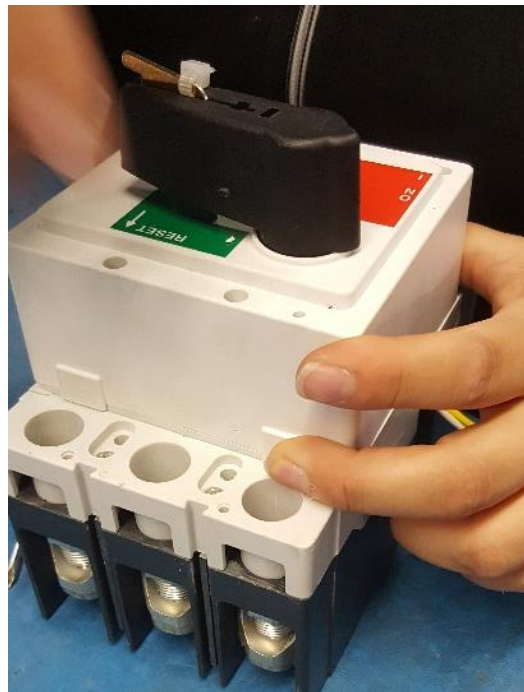
4.1.9 Replacement of CB1 Shunt Trip

If the circuit breaker shunt trip is deemed defective as per the Troubleshooting Guide in 4.1.16 follow the steps below to replace the circuit breaker.

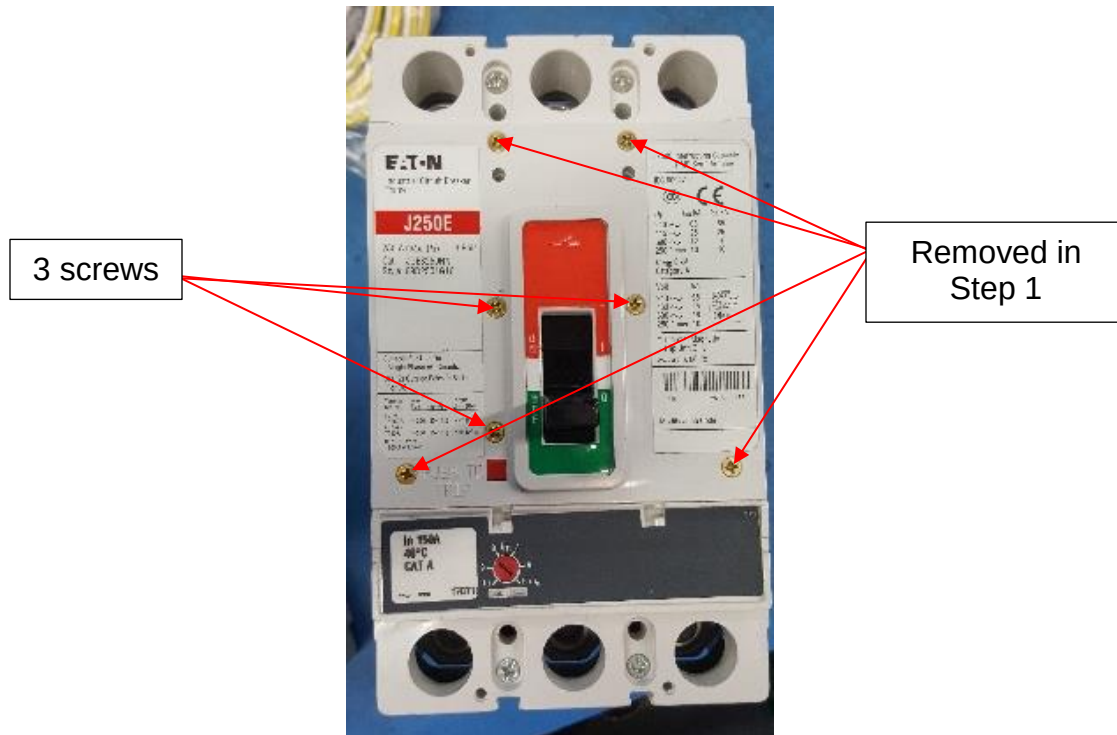
1. On the front of CB1, remove the 4 screws holding the circuit breaker handle in place.



2. Remove the handle from the circuit breaker.



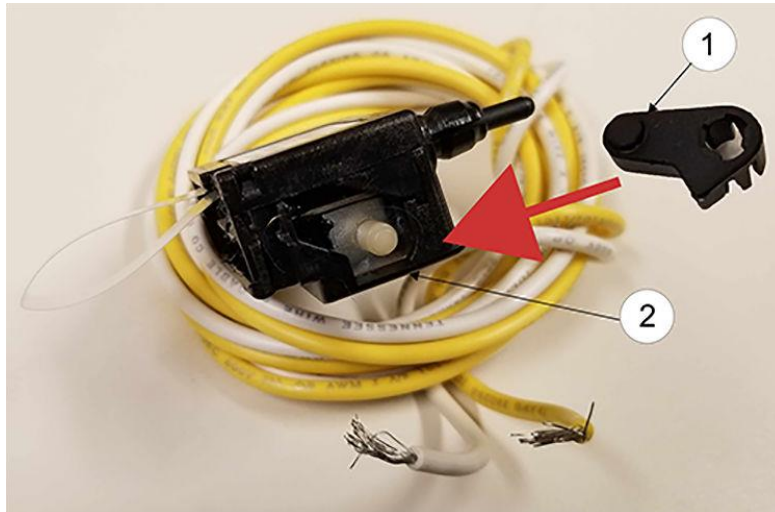
3. Remove the remaining 3 screws to remove the circuit breaker cover.



4. Remove the shunt trip from the left side of the circuit breaker by pulling straight up on the pull tab. Cut the shunt trip wires to fully remove the old shunt trip.



5. Insert the JG adapter on the shunt trip.



6. Insert the new shunt trip into the circuit breaker and route the wires. Butt splice the new shunt trip's wires with the old wires.

Gauge info - Butt Splice, #16-14 AWG insul. blue.

Wire	Voltage	Wire size
Yellow	120V	14 AWG
White	120V	14 AWG

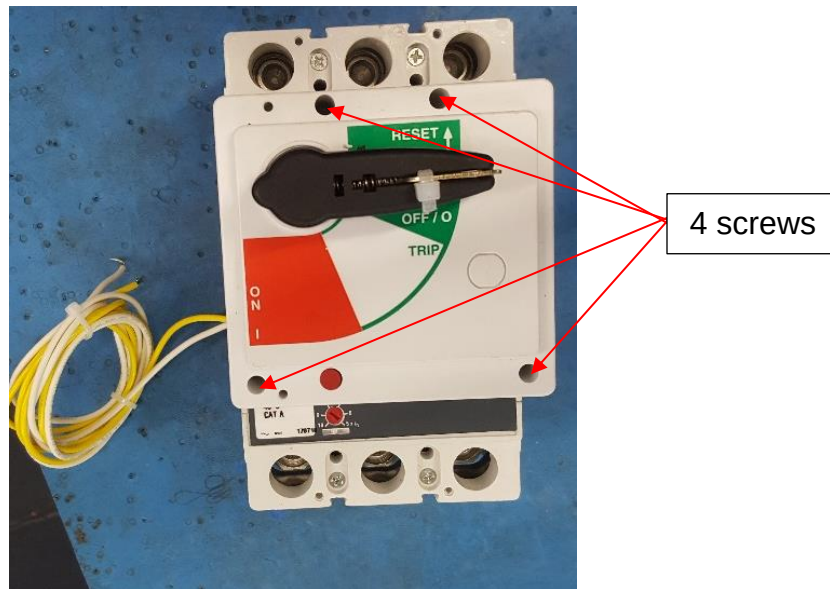




7. Reattach the circuit breaker cover using ONLY the 3 screws removed in step 3.



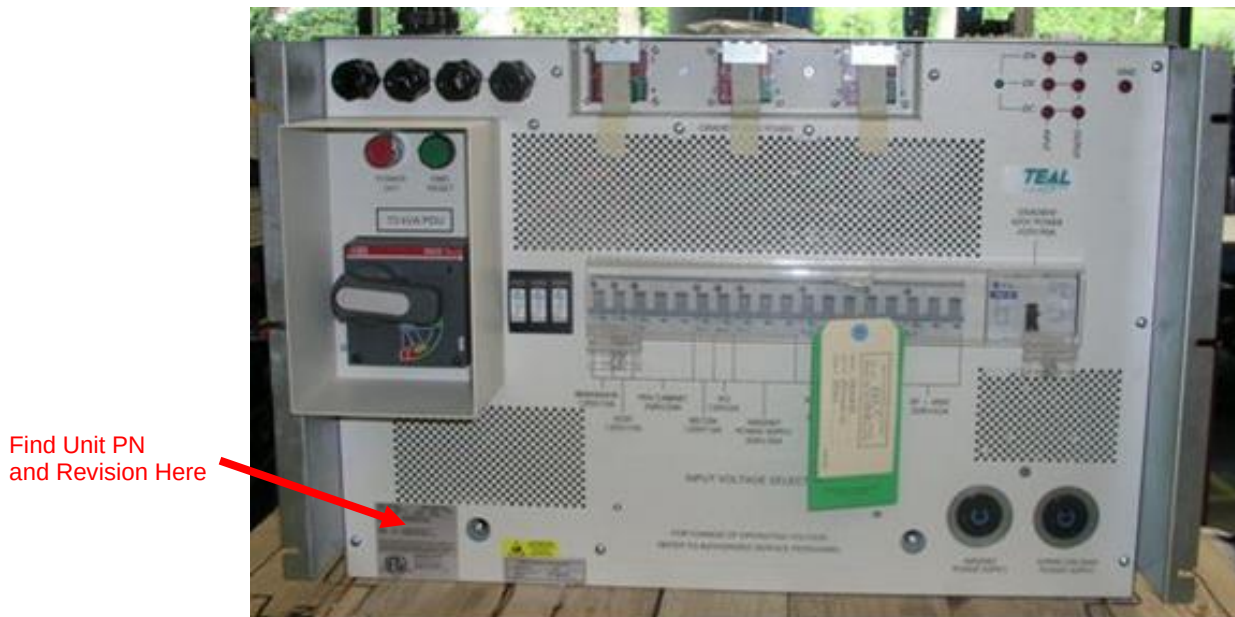
8. Place the handle removed in step 2 over the circuit breaker and secure using the remaining 4 screws removed in step 1.



9. Test the full range of the handle.

4.1.10 Replacement of Gradient 420V Power Circuit Breaker (CB2)

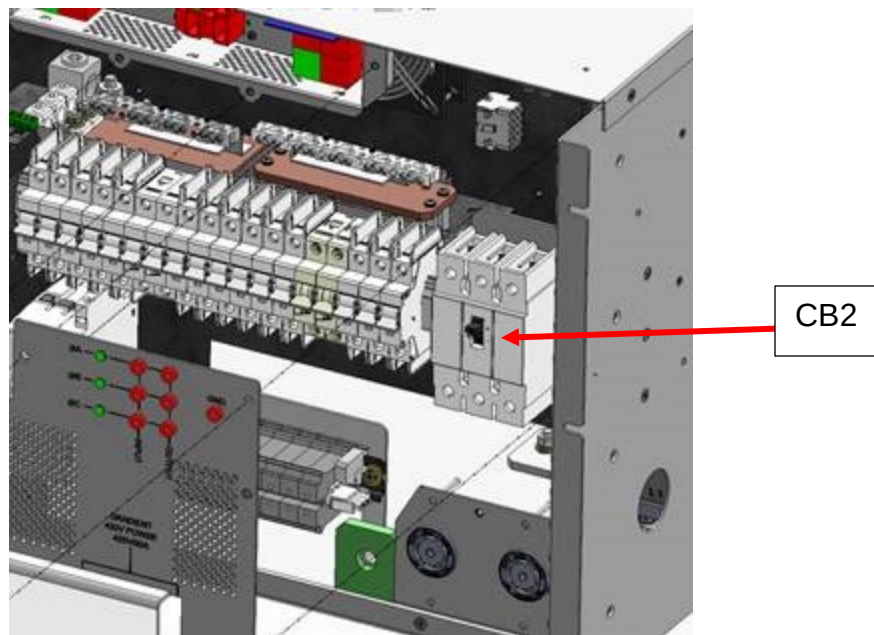
Before moving forward with circuit breaker replacement, you will need to identify the unit part number (PN) and revision of the unit being worked on. Proceed to the appropriate section below that corresponds with the unit.



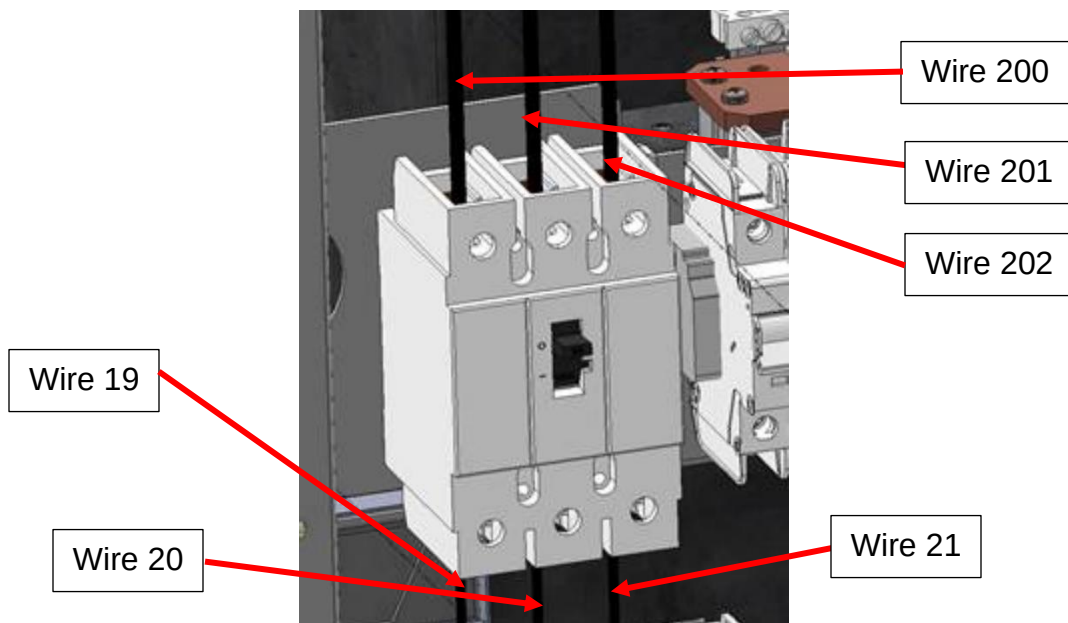
If the circuit breaker is deemed defective as per the Troubleshooting Guide in 4.1.16 follow the steps below to replace the circuit breaker.

Tools Needed: Phillips and flathead screwdriver, open wrench

1. Remove the front panel per section 4.1.2.



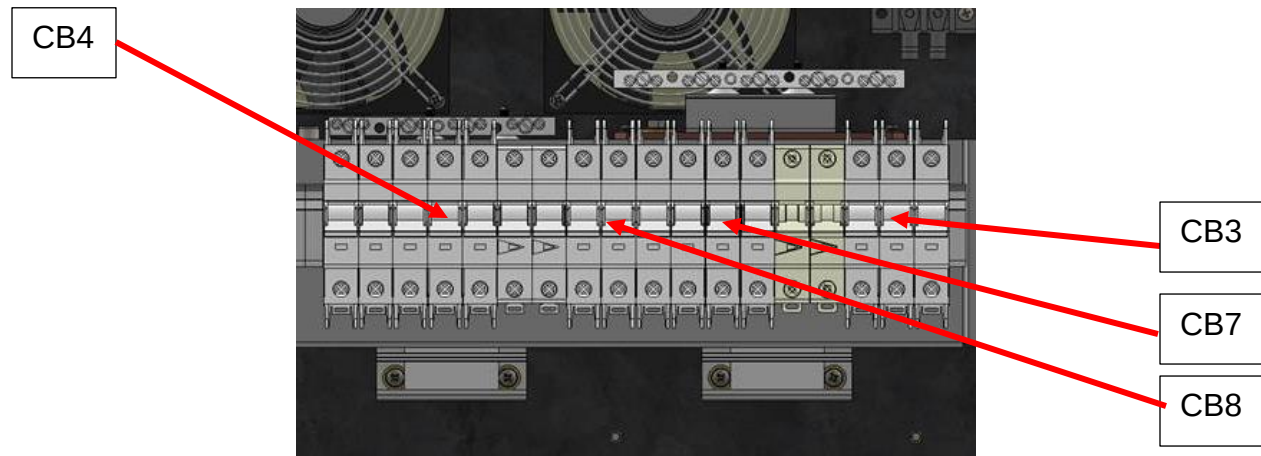
2. Disconnect wires 19, 20 and 21 from the input and 200, 201, 202 from the output of the Eaton circuit breaker CB2 using a flathead screwdriver. Label wires with masking tape if necessary.



3. Remove the 4 mounting screws holding the circuit breaker to the plate using a flathead screwdriver.
4. Remove the circuit breaker from the mounting plate.
5. Install the new Eaton circuit breaker in place (4 screws).
6. Reconnect the wires removed in step 2.

4.1.11 Replacement of RF-MNS, Pen Cabinet, Supercon Shim PS, and Magnet Power Supply Circuit Breakers (CB3, CB4, CB7, CB8)

1. Remove the front panel per section 4.1.2.
2. Disconnect wires 151, 152/159, and 153 from the input (bottom) of the Eaton circuit breaker CB3. Disconnect wires B-CB3, C-CB3, and A-CB3 from the output (top) of the circuit breaker. Label wires with masking tape if necessary.



3. Remove the circuit breaker from the DIN rail by gently pulling on the tab on the bottom of the circuit breaker using a flathead screwdriver.
4. Install the new Eaton circuit breaker on the DIN rail.
5. Repeat steps 2 through 4 for CB4, CB7, and CB8. Use Table 4.1.11.1 for the proper wire numbers.

Table 4.1.11.1

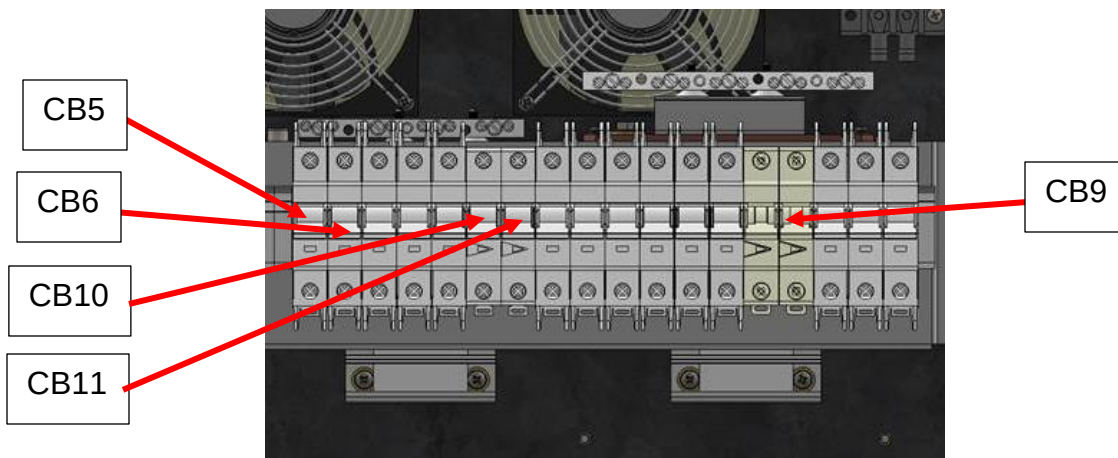
Circuit Breaker	Input Wire Number	Output Wire Number
CB4	154, 155, 156	CB4-1, CB4-2, CB4-3
CB7	134, 135, 136	140, 141, 142
CB8	131/134, 132/135, 133/136	137, 138, 139

6. Reconnect the wiring removed in step 2.

4.1.12 Replacement of Brainwave, Host, Gradient 48V Control, Recon, and SIO Circuit Breakers (CB5, CB6, CB9-CB11)

Tools Needed: Phillips screwdriver, open wrench

1. Remove the front panel per section 4.1.2.
2. Disconnect wire 210 from the input (bottom) of the Eaton circuit breaker CB5. Disconnect wire CB5 from the output (top) of the circuit breaker. Label wires with masking tape if necessary.



3. Remove the circuit breaker from the DIN rail by gently pulling on the tab on the bottom of the circuit breaker using a flathead screwdriver.
4. Install the new Eaton circuit breaker on the DIN rail.
5. Repeat steps 2 through 4 for CB6, CB9, CB10, and CB11. Use Table 4.1.12.1 for the proper wire numbers.

Table 4.1.12.1

Circuit Breaker	Input Wire Number	Output Wire Number
CB6	64	CB6
CB9	26, 27	89, 90
CB10	157	158
CB11	159	160

6. Reconnect the wiring removed in step 2.

4.1.13 Replacement of Receptacles and Terminal Blocks (REC1-REC3, TB1-TB3, TB6)

Access to some FRU's might be difficult. Additional components within the cabinet will need to be removed and/or PDU removed from the cabinet in order to access the FRU if replacement is desired.

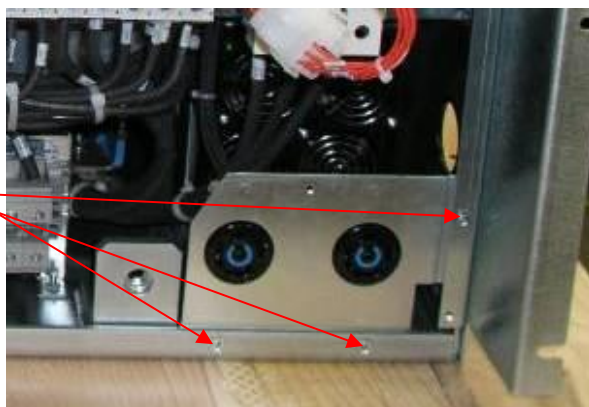
Tools Needed: Phillips and flathead screwdriver, open wrench

1. Remove the front panel per section 4.1.2.

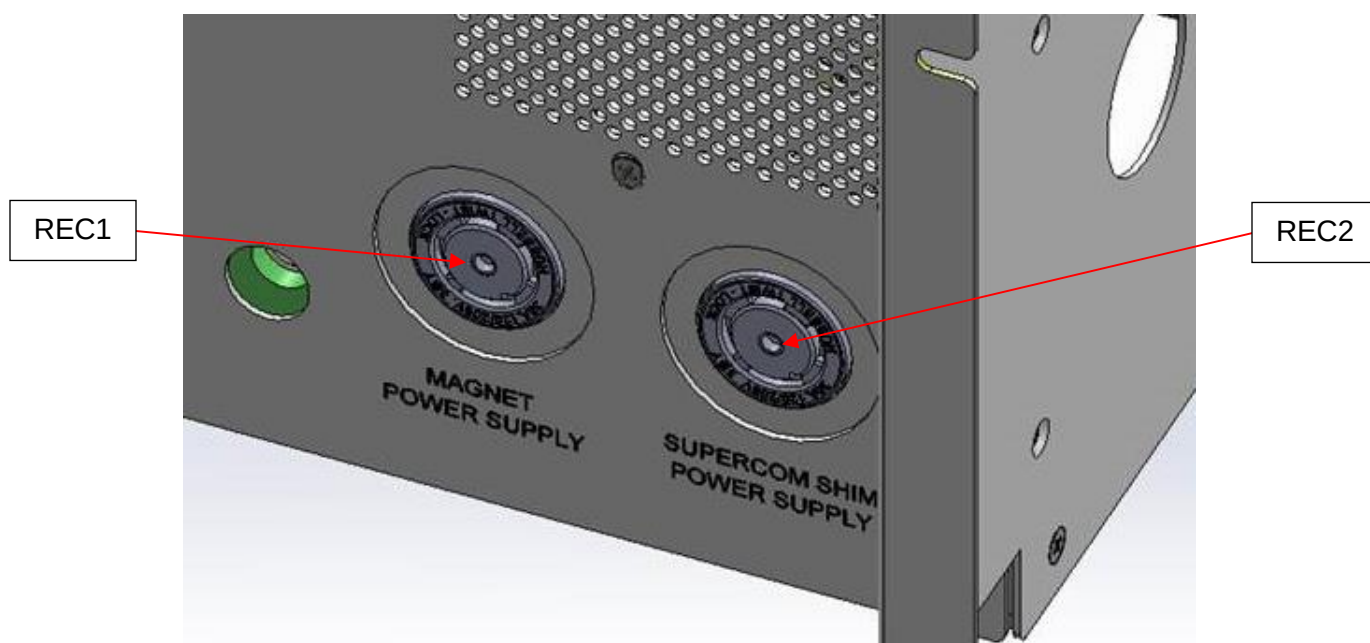
4.1.13.1 Replacement of REC1 and REC2

1. Remove the 3 (10-32x1/2) screws at the receptacle assembly.

3 (10-32x1/2)
screws



2. Disconnect wires 137 (X), 138 (Y), 139 (Z), 144 (W), and 143 (G) from REC1. Label wires with masking tape if necessary.



3. Remove the screws holding REC1 in place.
4. Repeat for REC2 with wires 140 (X), 141 (Y), 142 (Z), 144/146 (W), 1433/145 (G). Label wires with masking tape if necessary.
5. Install the new REC1 and REC2.

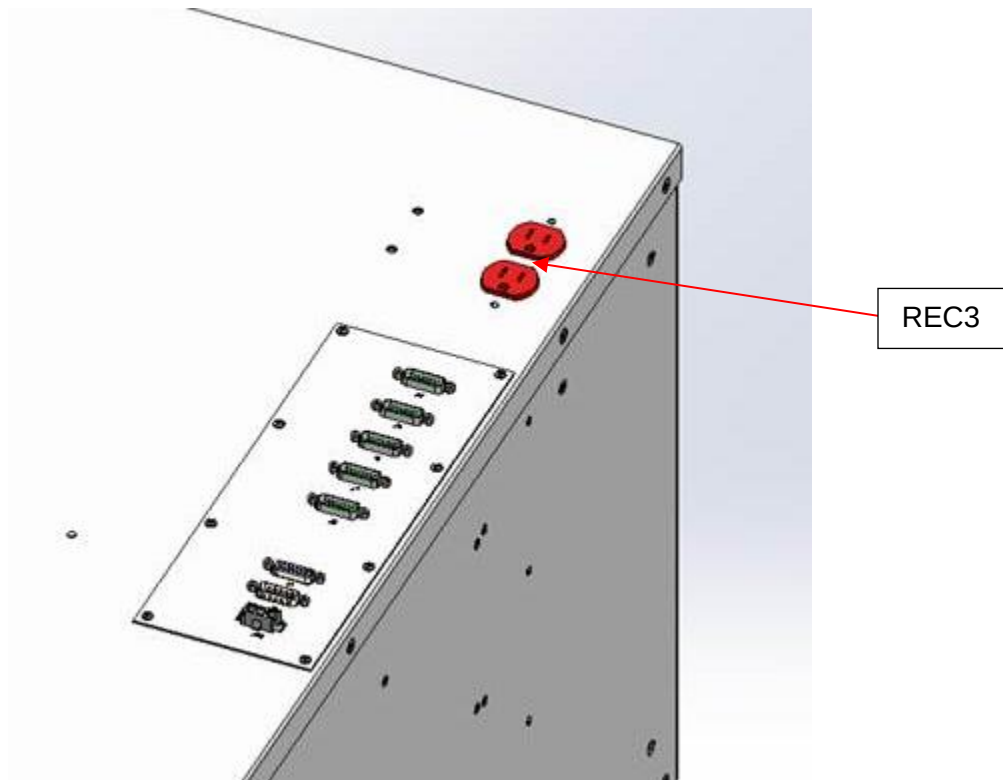
4.1.13.2 Replacement of REC3

NOTE: Additional components within the cabinet will need to be removed and/or PDU removed from the cabinet in order to access the FRU if replacement is desired

1. Remove the 4 (10-32x3/4) screws from the back panel mounting bracket.
2. Remove the 14 (10-32x1/2) screws on the back panel and set aside.



3. Disconnect wires 158 and 160 (H), 161 and 162 (N), and 205 (G) from REC3. Label wires with masking tape if necessary.

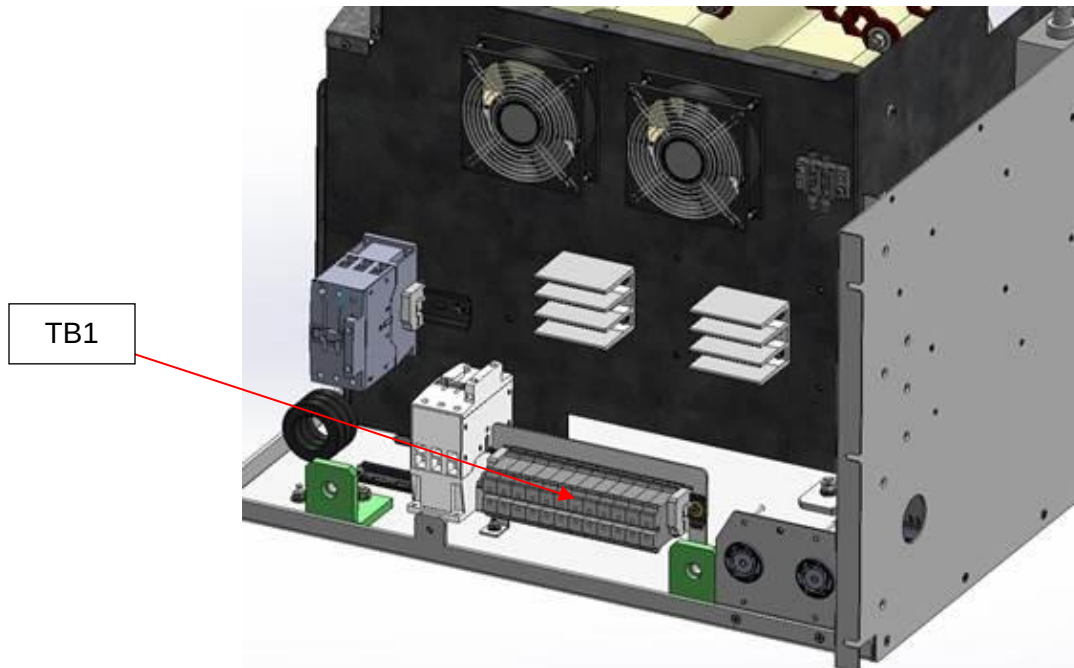


4. Remove the screws holding the receptacle in place.
5. Install the new receptacle.

4.1.13.3 Replacement of TB1

NOTE: It is recommended to only replace the TB slots used for the transformer taps to minimize risk of miswiring.

1. Identify the transformer tap configuration and note their positions along the terminal block.
2. Disconnect wires 4-6 from the top of TB1 using a flathead screwdriver. Disconnect the corresponding wire from the bottom of the 3 TB slots. Label wires with masking tape if necessary.

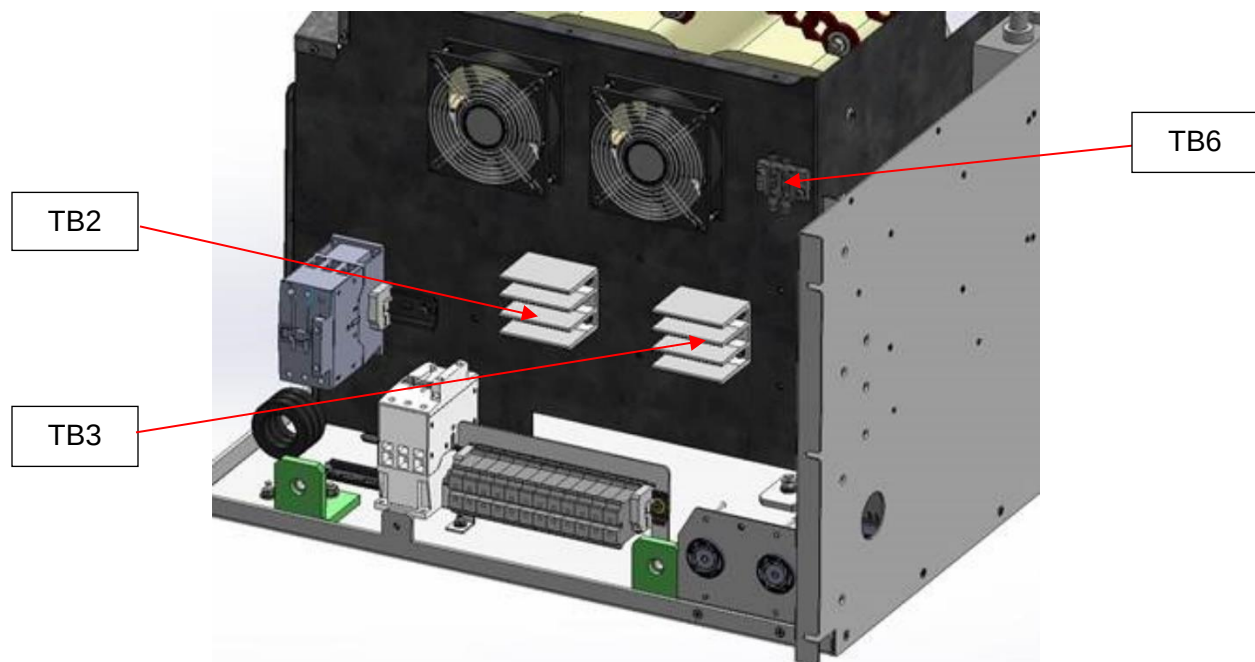


3. Remove the 3 TB slots by first removing one of the end caps from the DIN rail. Separate the 3 TBs slightly from the others and pry each off the DIN rail using a flathead screwdriver.
4. Install the new TB slots on the DIN rail.
5. Reconnect the wires removed in step 2.

4.1.13.4 Replacement of TB2, TB3, and TB6

NOTE: If replacement of TB2 and TB3 is desired, access is difficult unless the panel of mini CBs is removed.

1. Disconnect wires 24, 28/59, and 29/60 from the input (left) of TB2. Label wires with masking tape if necessary.
2. Remove the screws holding TB2 in place.



3. Remove TB2 from the panel.
4. Install new TB2.
5. Reconnect the wires removed in step 1.
6. Repeat steps 1 through 3 for TB3 and TB6. Use Table 4.1.13.4.1 for the proper wire numbers.

Table 4.1.13.4.1

Terminal Block	Input Wire Numbers	Output Wire Numbers
TB3	128, 129, 130	131/210, 132/157, 133
TB6	57, 58	62, 25/62

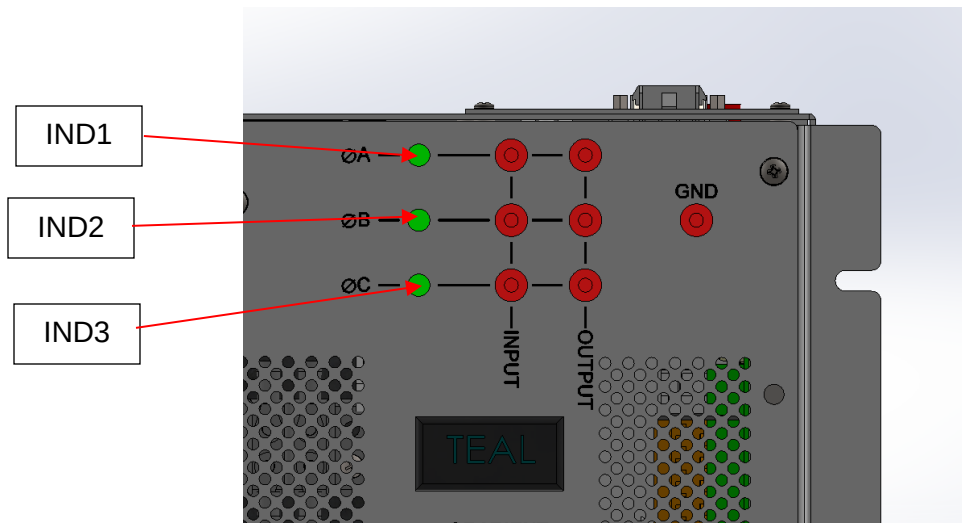
4.1.14 Replacement of Indicators, Switches, and Test Points (IND1-IND3, SW1, SW2, TP1-7)

Tools Needed: Phillips screwdriver, open wrench

1. Remove the front panel per section 4.1.2.

4.1.14.1 Replacement of Indicators IND1-IND3

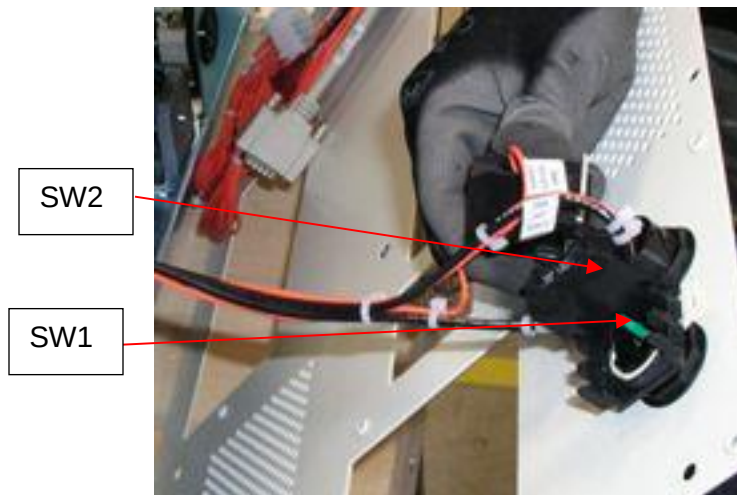
1. From the back of the front panel, remove wires 192 and 193 from IND1, wires 224 and 225 from IND2, and wires 226 and 227 from IND3 by pulling them from the 15-pin connector.



2. Pull out the 3 indicator assemblies from the front panel.
3. Install the new indicator assemblies.

4.1.14.2 Replacement of Switches SW1 and SW2

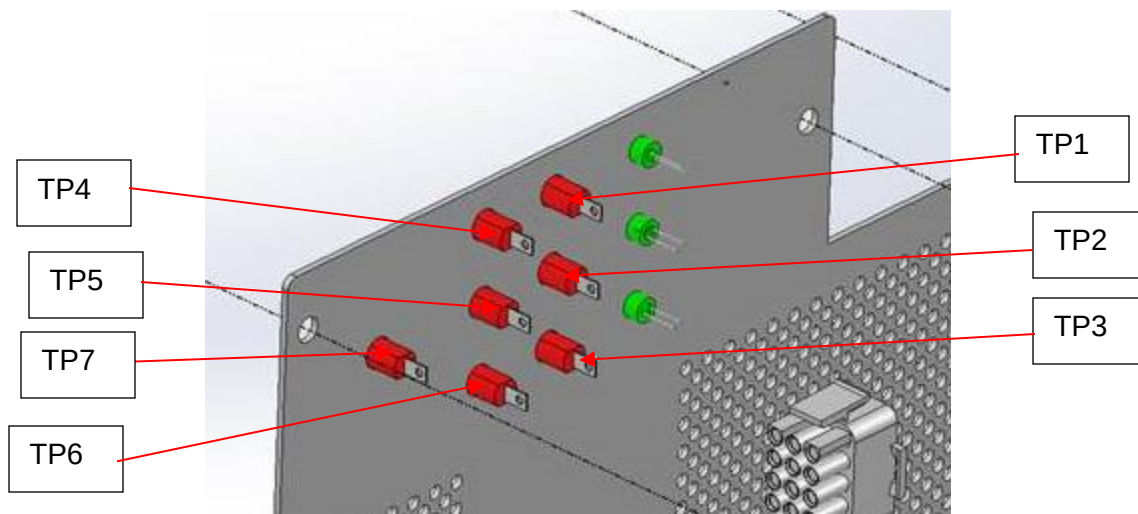
1. Disconnect wires 55/71 and 56/72 from SW1 and wires 74 and 75 from SW2.



2. Disassemble SW1 to remove it from the front panel by unscrewing the nut on the back of the switch. Repeat for SW2.
3. Install the new SW1 and SW2 in place.
4. Connect the wiring removed in step 1.

4.1.14.3 Replacement of Test Points TP1-TP7

1. Disconnect wire 185 from TP1.



2. Remove the nut on the back of TP1 and slide TP1 out of the front panel.
3. Install the new TP1 and connect the wire removed in step 1.
4. Repeat for TP2 through TP7 using Table 4.1.12.3.1 for the proper wire numbers

Table 4.1.12.3.1

Test Point	Wire Number
TP2	186
TP3	187
TP4 (Ground)	188
TP5	189
TP6	190
TP7	191

4.1.15 Placement of New FRUs

1. To replace the FRUs in Sections 4.1.3 through 4.1.14 following the instructions in reverse order.
2. Use the schematic as needed to ensure placement of wires and/or cables.
3. To close the unit back up following the directions of Section 4.1.2 in reverse order.
4. Once unit has been closed up, turn on Main Input circuit breaker.

4.1.16 Troubleshooting Guide

The following table lists fault conditions that may be encountered along with possible causes and solutions.

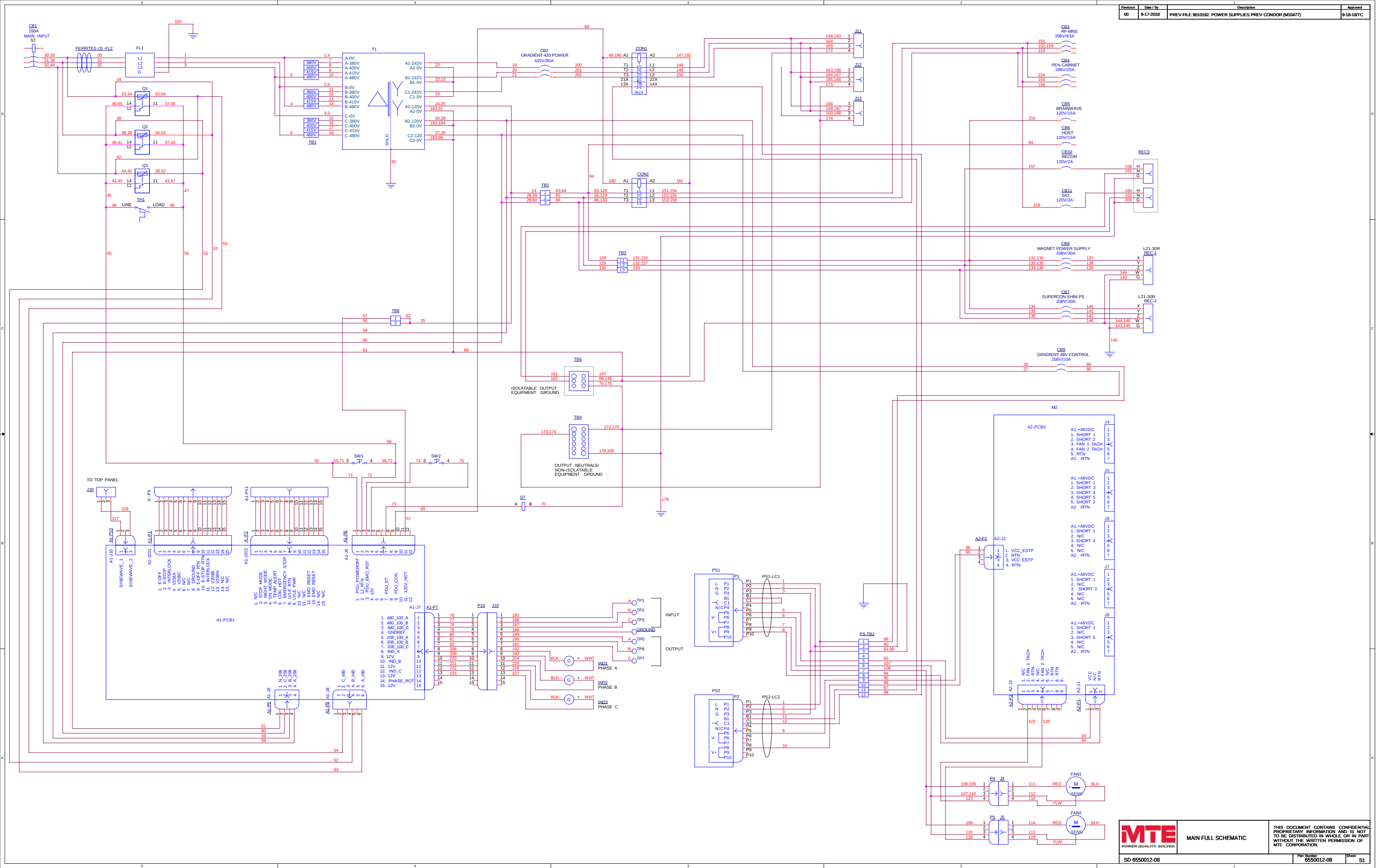
Troubleshoot Guide

Table 4-2

MODULE	SYMPTOM	CAUSE	SOLUTION
PDU	Main Input CB Trips	Over-temperature	1. Investigate cause of overtemperature 2. Allow unit to cool down. 3. Reset the circuit breaker
		Shorted Power Off Switch	1. Turn power off 2. Reset Main Circuit Breaker 3. Turn power on 4. If CB trips, replace PDU
		AUX Module LEAK Sensor Signal	1. See "Leak Detection Sensor Theory & Troubleshooting" service methods document under troubleshooting folder
	Distribution Breakers Tripping	Defective Circuit Breaker	1. Turn power off 2. Check if circuit breaker still trips (won't latch up)
		Overload	1. Check load by disconnecting cabinet from PDU whose circuit breaker is tripping, or use an Amp Clamp / DVM to check current draw. 2. Correct load problem.
	CON1/CON2 Contactor not operating	Defective Contactor	1. Ensure that the contactor CON1 is receiving 120V to the coils. 2. If 120V is present, replace the contactor. 3. If 120V is not present, verify that CON2 is receiving 120V to the coils 4. If 120V is present, replace the contactor. 5. If 120V is not present, replace the PDU.
		Defective Control Board A1	1. Connect test plug and verify PDU operation. 2. If PDU operates with Test Plug, then replace the cable that connects to J2. 3. If PDU does not operate with Test Plug, then replace Control Board A1
	Indicators not lighting when power is applied	Input Circuit Breaker off	1. Turn Input Circuit Breaker, CB1, on
		Input Phase Missing	1. Check that other phase LED's are on 2. Measure at TP
		Light Burned Out	1. Check that other phase LED's are on. 2. Replace PDU
		Not Getting Proper Output	1. Measure input and output, at front panel test points, of each phase, from phase to phase (208 VAC: 208 VAC) and phase to ground (120 VAC: 120 VAC) 2. If difference is not a 100:1 ratio, replace PDU
	Unit does not exit ESTOP mode (cannot turn on output contactor)	Control Board not receiving reset signal	1. Try resetting the unit by pressing SW1 on the front panel
		Control Board not receiving power	1. Confirm connection on the board 2. Confirm board is receiving 120V by measuring at P6-11(hot) and P6-9 (neutral)
		Control Board switch in wrong position	1. If control board replaced before this issue then see section 4.1.3 of control board replacement and confirm correct switch and jumper locations

Troubleshoot Guide
Table 4-2 (cont.)

MODULE	SYMPTOM	CAUSE	SOLUTION
	Fans not running	CB9 is OFF	1. Ensure CB9 is in the ON position
		PS1 Power supply (upper unit) not receiving power	1. Ensure CB9 is in the ON position 2. With CB9 on, measure for 120VAC at PS1 J1-1 and PS1 J1-2
		Defective power supply	1. With CB9 on and 120V present at PS1 J1, look for the green LED to be lit on the front of the power supply 2. If the green LED is not lit, then the power supply is likely defective, replace the power supply 3. If the green LED is lit, then measure for 48VDC at the power supply output PS1 J7 4. If 48V is not present, the likely the power supply is defective, replace the power supply
		Defective Fan	1. Measure for 48VDC at the fan connectors P4-1 & P4-3 and P5-1 & P5-3 2. If 48VDC is present, then replace the fan
	Voltage at one or more output missing	Output breaker in the OFF position.	1. Turn ON all output breaker
		Unit is still in ESTOP mode	1. Remove E-STOP condition.
	Internal system AC line clock signal missing	Control Board Failed	1. Replace Control Board.
	Power supply LED is off with power applied and PDU is not in E-Stop mode	Defective power supply	1. Replace power supply
	Power supply LED is lit but has no 48V output	Defective power supply	1. Replace power supply



APPENDIX A

GE FIELD NOTIFICATION: Instructions to replace Condor 48V PS Assy. with Delta 48V PS assembly.

SUMMARY:

Due to frequent repairs on the 48V PDU power supplies in the XGD and XGD2 PDUs, the Condor supply, GE PN 5140621-4 will be replaced with the Delta supply, GE PN 5745311. When either of the current Condor PS1 or PS2 power supplies fail, both Condor power supplies will need to be replaced with (2) Delta power supplies and the new Delta compatible bracket. This change will require the field to remove the existing Condor power supplies and bracket, install the Delta power supply bracket (GE PN 5140621-10) and 2 new Delta power supplies (GE PN 5745311).

The Delta power supply bracket has been redesigned to allow 'blind mating' for more efficient future repairs (new part number 5343114-10) and will be phased into new production and service inventory once the existing supply of Delta power supply bracket 5140621-10 is diminished.



Notice:

The auxiliary contacts on the contactor (CON1), located on the left side of the PDU, will need to be switched from NC (normally closed) to NO (normally open) to activate the new Delta power supplies.

PARTS IN SCOPE:

Part Number	Description	Comments
5140621-4	Condor power supply	Recycle
5745311	Delta power supply	Replacement—requires (2) initially to replace (2) Condor supplies
5140621-10	Delta power supply bracket (existing)	Replacement – recycle Condor bracket, replace with Delta bracket.
5343114-10	Delta power supply bracket (blind-mate)	

OVERVIEW:

The following will need to be performed for the 48V PDU power supply replacement procedure:

1. LOTO PDU
2. Remove Condor power supplies
3. Replace with Delta power supplies
4. Switch AUX contacts on CON1 from NC to NO (required for Delta PS)

5. Remove LOTO at the MDP, power up the PDU, and perform *Check Scan*



Condor 48V supply in PDU

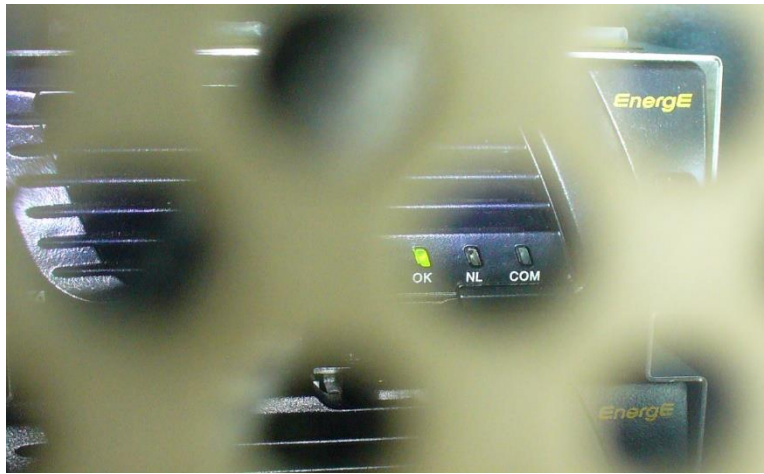
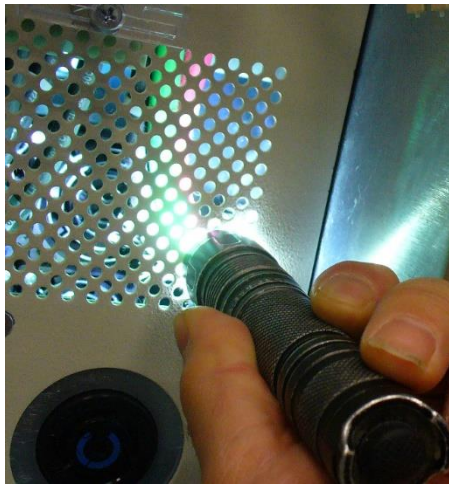


Delta 48V supply in PDU (existing bracket)



Delta 48V supply in PDU (blind-mate bracket)

Due to the difference in construction of the power supply bracket assembly, when one supply fails, both Condor supplies (5140621-4) must be replaced with the (2) Delta supplies (5745311) and (1) Delta power supply bracket (5140621-10). However, once the new bracket assembly is installed, if one Delta supply fails (5745311), then just that one Delta supply can be replaced. To determine if the site has Condor or Delta installed in the PDU, a maintenance sticker (5661793) will be placed on the front of the PDU. If the sticker is not visible, then using a flashlight, peer through the PDU vent holes, to see the name **Delta EnergyE**.



Power Supply LED Status:

LED	Status	Function
OK	On – Green (normal operation) Off Blinking	Operating mode Shut down mode Standby mode
NL (no load)	On – Yellow Off – (normal operation) Blinking	Output current <5% of nominal load Output current within 5% nominal load and current limit Output at current limit mode
COM (communication)	On – Green Off -- (normal operation)	Communication successful with PSC-3 No communication with controller, bus cabling faulty, bus termination faulty.

RESOLUTION: Replacement of Condor 48V Power Supplies to Delta 48V Delta Power Supplies

Required Tools:

- Phillips screwdriver or power screwdriver set
- 7/16" open wrench
- Small flat-head screwdriver

1. Removal of the Front Panel

Turn off the main input circuit breaker by rotating the lever to the off position. Perform the lockout-tagout (LOTO) procedure on the PDU main switch at the MDP.

Note: *There are slight differences in the XGD and the XGD2 PDUs:*

XGD

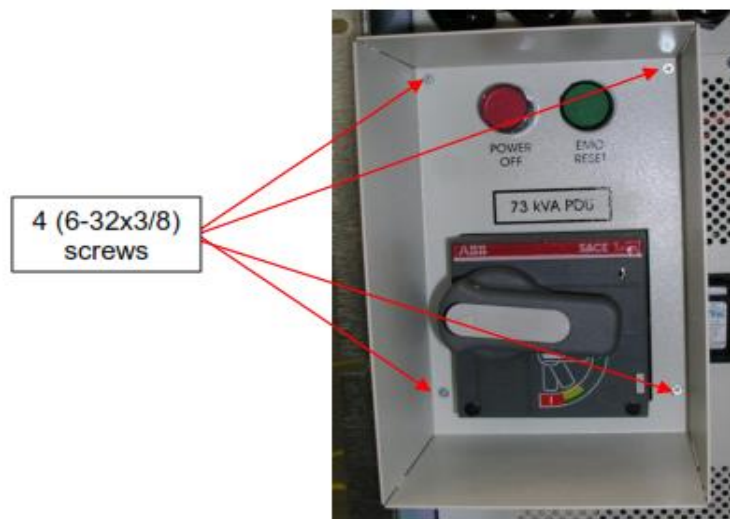


XGD2



2. For both models, remove the 4 (6-32x3/8) screws on the circuit breaker guard using a Phillips screwdriver.

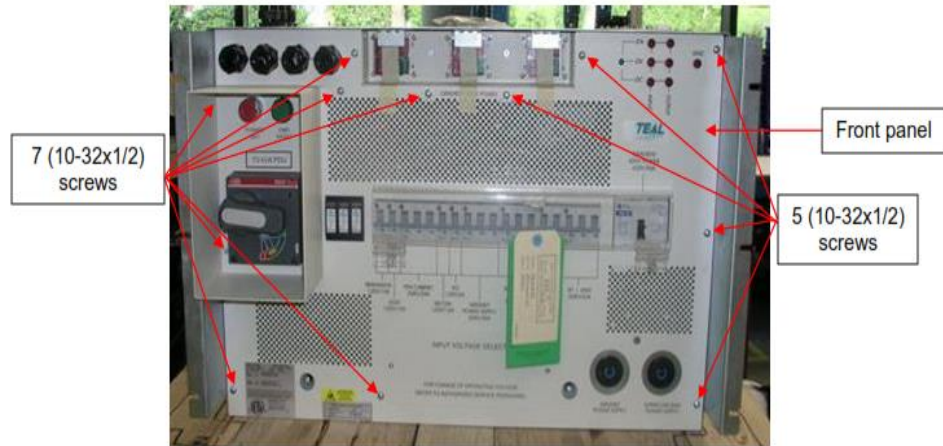
Note: *Do not recycle the screws, as they will be used to reattach the guard back over the circuit breaker.*



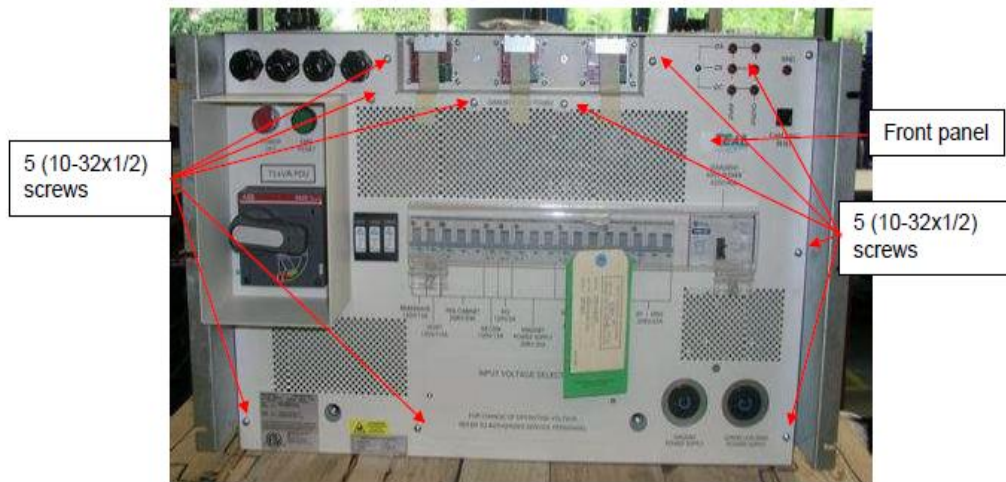
3. Remove the (10-32x1/2) screws on the front panel using a Phillips screwdriver.

Note: *Do not recycle the screws, as they will be used to reattach the front panel.*

XGD



XGD2



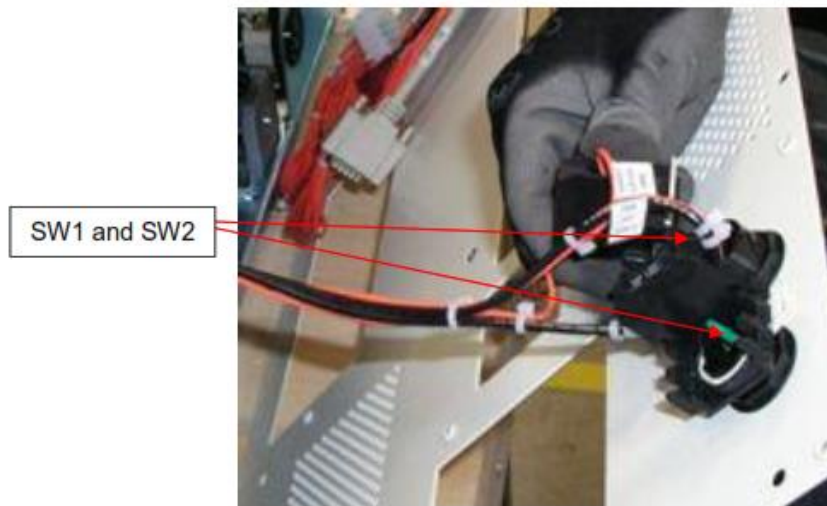
4. Slightly pull on the front panel and lean it down and away from the unit.

Note: Be careful not to damage the leak sensor strip.



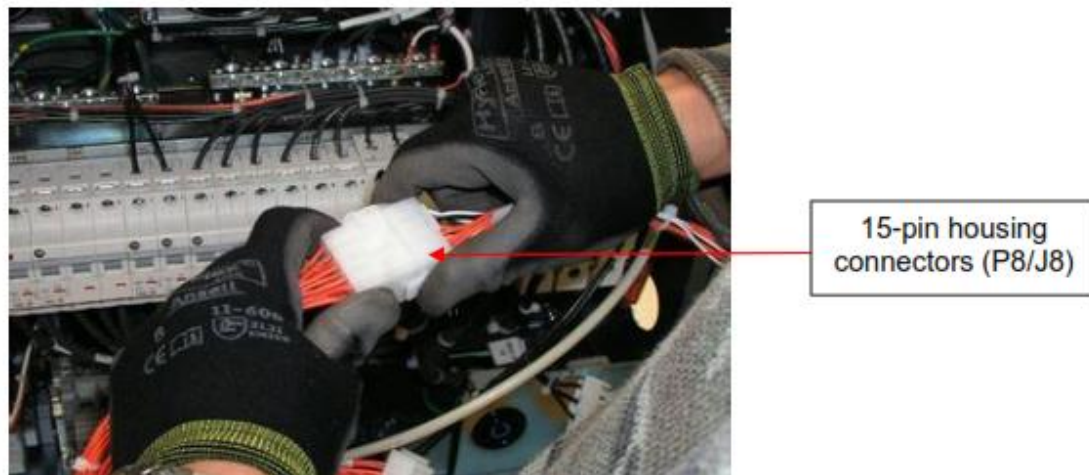
5. For both models, disconnect SW1 and SW2 from the back of the front panel.

Note: No tools are required. Turn tabs counterclockwise and gently pull the assembly apart.



6. For both models, disconnect the P8 and J8 15-pin housing connectors.

Note: No tools are required. Press down on the locking tabs and pull the connectors apart.



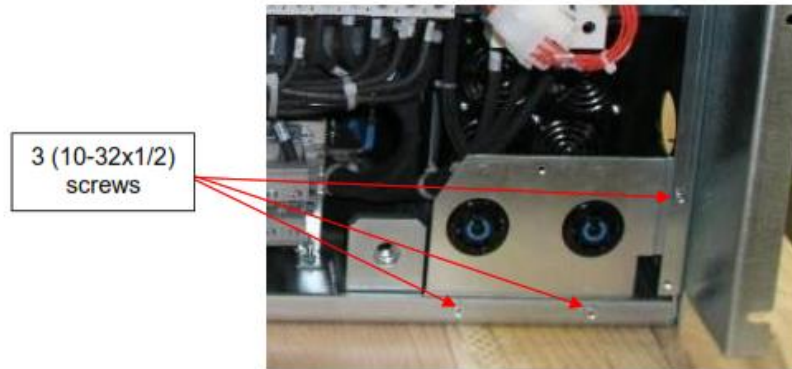
Replacement of Power Supplies (PS1 & PS2)

If the power supplies are deemed defective, follow the steps below to replace the power supplies. The power supplies are mounted behind the receptacle assembly of the PDU located on the bottom of the unit.

Note: When replacing a defective Condor power supply with a Delta power supply, a new Delta compatible bracket must be installed, and both Condor power supplies must be replaced.

1. For both XGD and XGD2, remove the 3 (10-32x1/2) screws at the receptacle assembly. Move the assembly out of the way.

Note: Do not recycle the screws, as they will be used to reattach the receptacle assembly.



! Note: Use Masking Tape and Marker to mark these wires to avoid mis-wiring after PS is replaced.

2. If more space is necessary, mark wires attached to the circuit breaker (s), then disconnect the wires.



3. Alternatively, remove the mounting screws from the circuit breaker (s) and move the circuit breaker to the side.

Note: Do not recycle screws, as they will be used to replace the circuit breaker.

XGD



XGD2

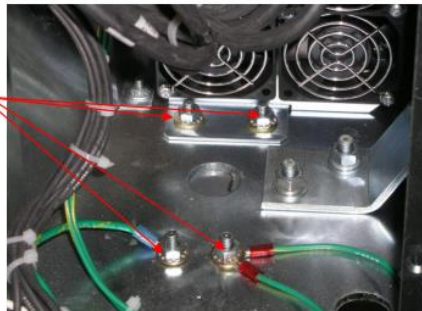
4 (8-32x3)
screws



4. Remove the 2 Kep nuts and 2 washers at the power supply assembly. Remove the 2 Kep nuts and 2 washers at the ground wires (if necessary, use a cutter to the tie wrap to allow slack for the power supply harness).

Note: Do not recycle the nuts and washers, as they will be used to reattach the power supply assembly and ground wires.

4 Kep nuts &
4 washers



Cut here

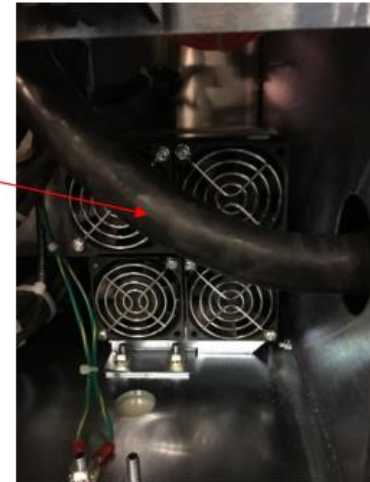


5. Carefully pull out the power supply assembly by lifting the assembly over the studs and sliding forward out of the front from the unit.

Note: Make sure to move the RF cable out of the way so as not to damage the outer jacket with the power supply housing.



RF cable



6. Mark the wire Run numbers / description on the wires with masking tape and a marker Disconnect ALL wires from the Condor power supply assembly.



Note: Proper marking of the wire numbers is critical when you reconnect to the new power supply, per wiring chart below.



Wiring Chart

From Condor Terminal Block	Run Marked on Wire	Description	To Delta Terminal Block Number
PS1 J1 #1	89	L AC Input Line	1
PS1 J1 #2	90	N AC Input Neutral	2
PS1 J1 #3	91, 99	G AC Input Ground	3
No Connection	No Connection	No Connection	4
PS1 J7 #1	93	PS1- Return	5
PS1 J7 #2	107	PS1- Return	6
PS1 J7 #3	108	PS1+ 48V	7

PS1 J7 #4	94	PS1+ 48V	8
PS2 J7 #1	95	PS2- Return	9
PS2 J7 #3	96	PS2+ 48V	10
PS2 J6 #3	98	B1 Common	11
PS2 J6 #4	97	C1 Inhibit	12

7. Attach the marked Condor wires to the terminal strip on the Delta bracket – note the Delta labeling shows the run number and the terminal block number with description (pin # 1 is on the left)

Delta PS Bracket (existing) |

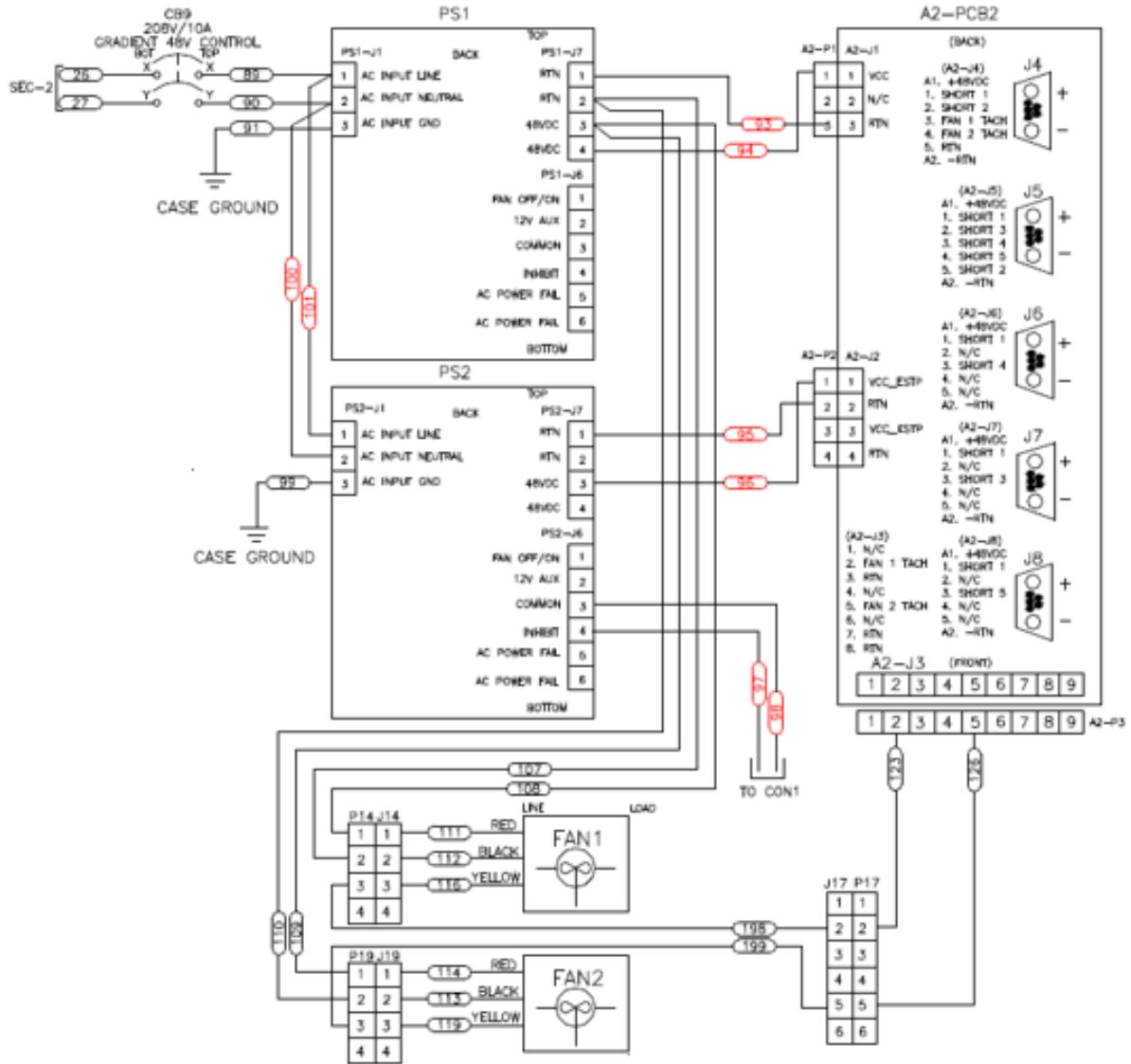


Delta PS Bracket (blind-mate)

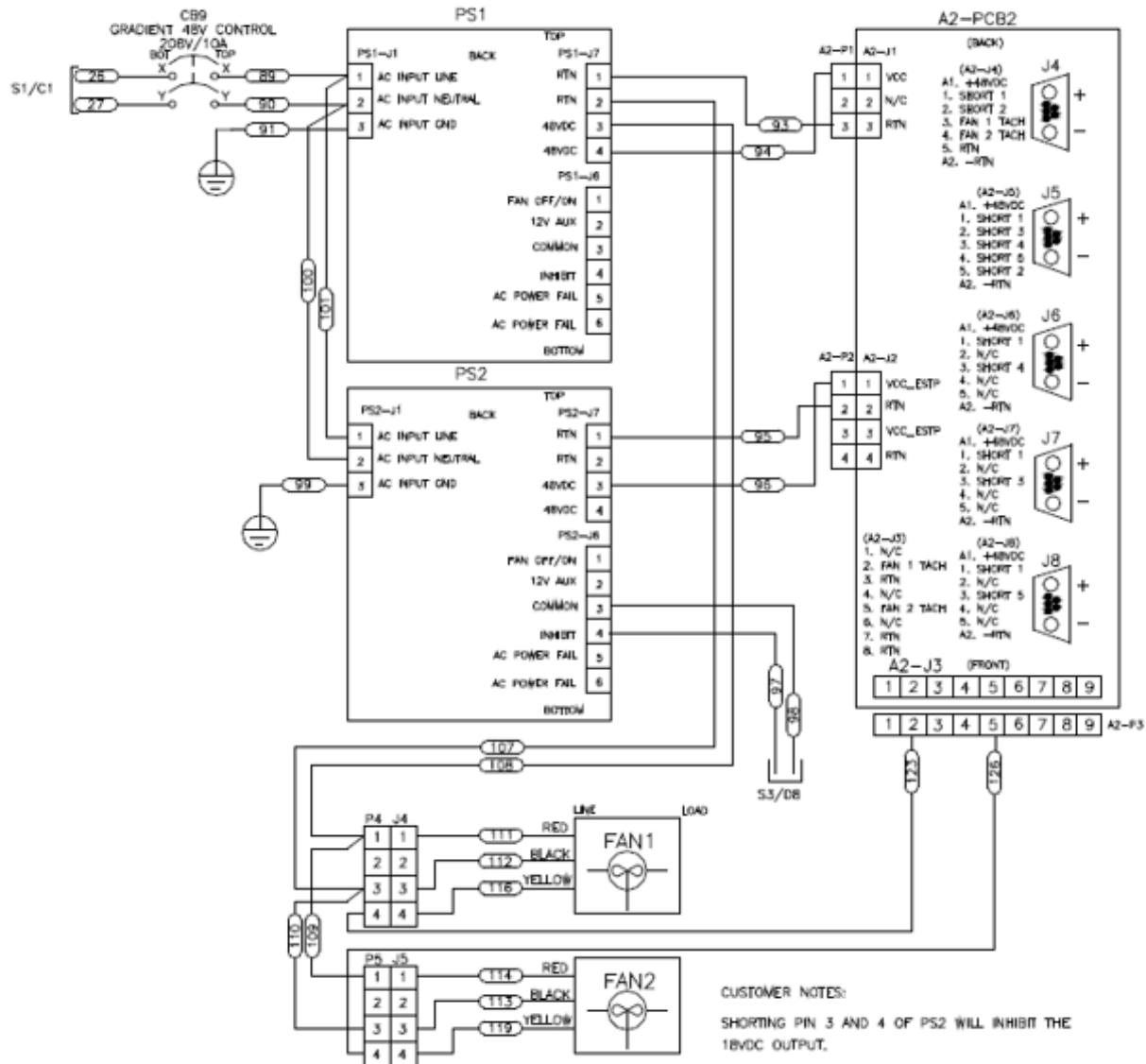


Note: XGD and XGD2 wiring diagrams are listed below.

XGD Power Supply Wiring Diagram



XGD2 Power Supply Wiring Diagram



For the existing Delta bracket:

8. Assemble the Delta power supply assembly using (1) bracket, FRU 5140621-10 and (2) Delta power supplies, FRU 5745311. Loosen the bottom three screws on one side of the bracket.

Note: Do not remove screws from the bracket.

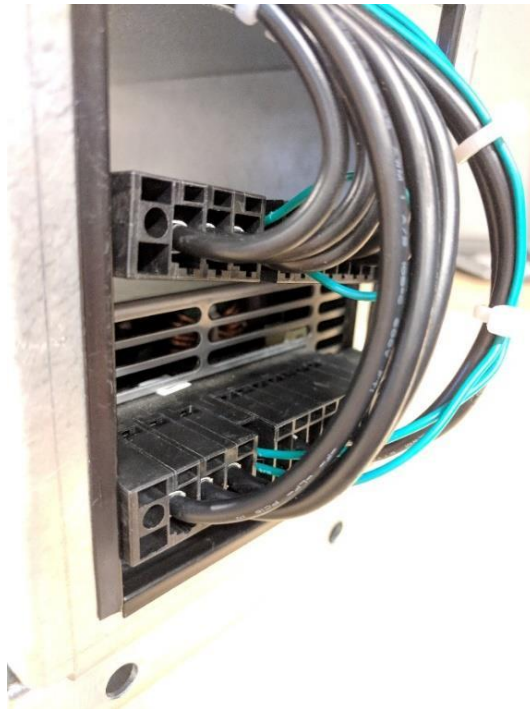


9. Begin sliding the first power supply PS2 (FRU 5745311) into the bottom slot of the bracket.



10. Attach the PS2 connector to the rear of the power supply (PS2). Once connected, slide the power supply (PS2) until it is against the back of the bracket.

Note: the harness is specifically manufactured such that PS2 harness will only reach the PS2 supply.



11. Latch the mounting pins on the power supply (PS2) through the locating hole on each side of the bracket.



12. Slide the second power supply PS1 (FRU 5745311) into the top slot of the bracket.



13. Attach the PS1 connector to the rear of the power supply (PS1).

Note: the harness is specifically manufactured such that PS1 harness will only reach the PS1 supply.



14. Once connected, slide the power supply (PS1) until it is against the back of the bracket.



15. Tighten the bottom (3) screws on the side of the bracket and latch the mounting pins on the power supply (PS1) through the locating hole on each side of the bracket (same as step 17 above). The FRU kit is now fully assembled.



For the Delta blind-mate bracket:

16. Gently slide the Delta power supplies into the bracket (FRU 5343114-10) to engage the rear blind-mate connector.



17. Lock power supplies into place using tabs on the front left / right of each supply.



Notice:

This is a critical step for functional operation, review the wiring diagrams if there are any questions.

18. Connect the wires that were removed and marked from the old power supply assembly and connect them to the terminal block on the new power supply assembly per the wire chart.

Delta PS Bracket (existing)



Delta PS Bracket (blind-mate)



Wiring Chart

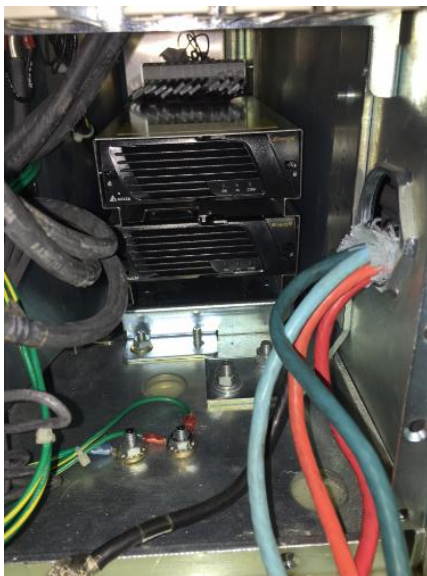
From Condor Terminal Block	Run Marked on Wire	Description	To Delta Terminal Block Number
PS1 J1 #1	89	L AC Input Line	1
PS1 J1 #2	90	N AC Input Neutral	2
PS1 J1 #3	91, 99	G AC Input Ground	3
No Connection	No Connection	No Connection	4
PS1 J7 #1	93	PS1- Return	5
PS1 J7 #2	107	PS1- Return	6
PS1 J7 #3	108	PS1+ 48V	7
PS1 J7 #4	94	PS1+ 48V	8
PS2 J7 #1	95	PS2- Return	9
PS2 J7 #3	96	PS2+ 48V	10
PS2 J6 #3	98	B1 Common	11
PS2 J6 #4	97	C1 Inhibit	12

19. Tilt the power supply assembly so that the terminal block clears the CB2 bracket.



20. Slide assembly into place where the old assembly was removed. Move the cables out of the way so as not to damage the outer jacket with the power supply housing.

XGD

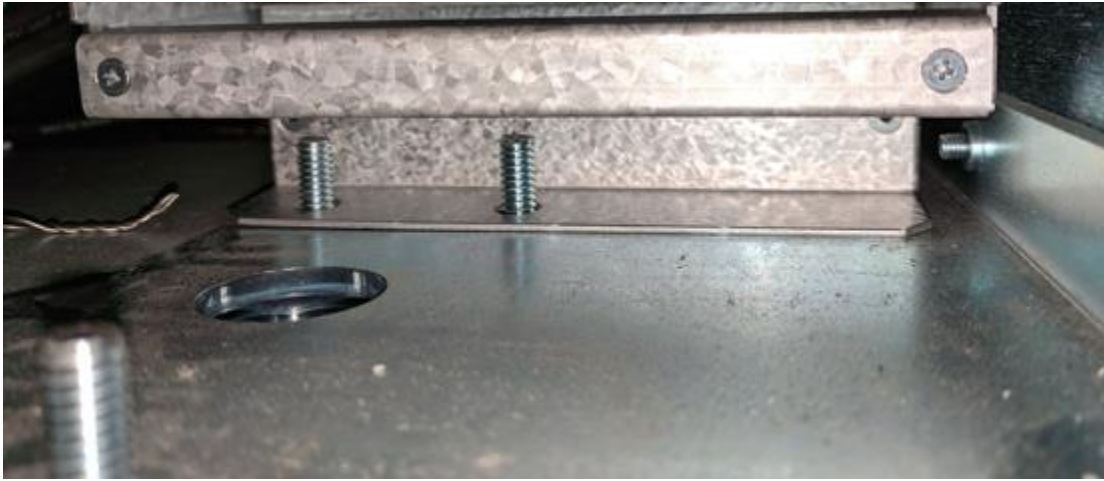


XGD2

RF cable



21. Reinstall the mounting hardware (2 Kep nuts and 2 washers) on the power supply assembly bracket and (2 Kep nuts and 2 washers) on the ground wire connections that were disconnected in step 10.

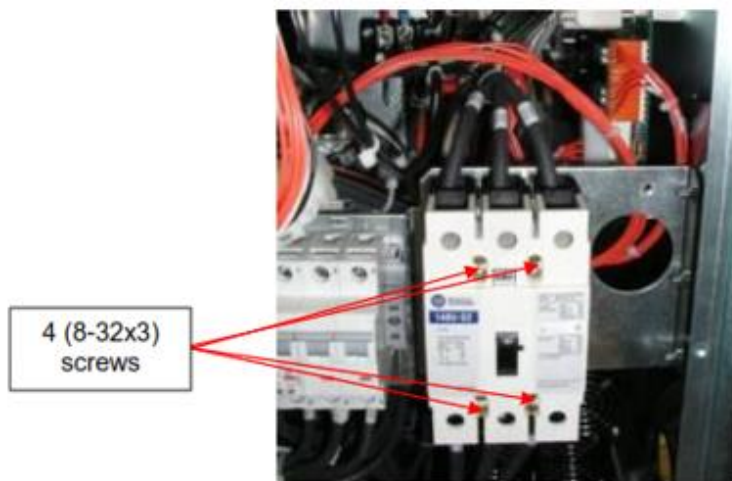


22. Reinstall CB2 (4 screws), if it was removed in step 9.

XGD



XGD2

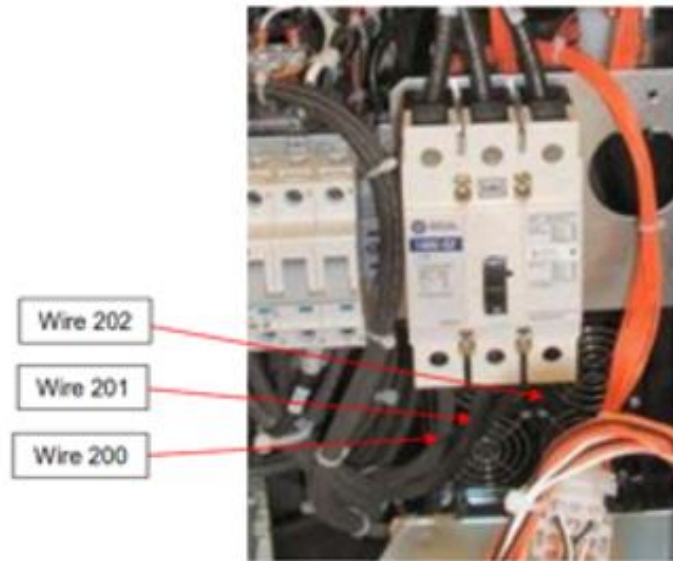


23. Reconnect wires 200, 201, and 202 to the bottom of CB2, if disconnected in step 8.

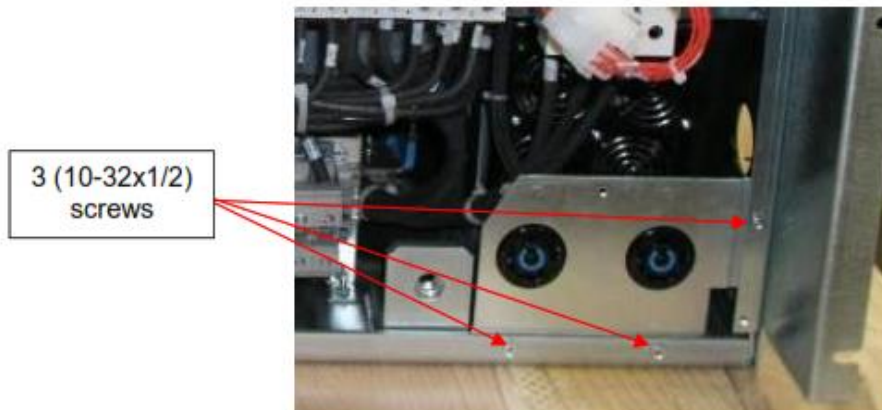
XGD



XGD2



24. Reinstall the receptacle assembly (3 screws), if removed in step 7.

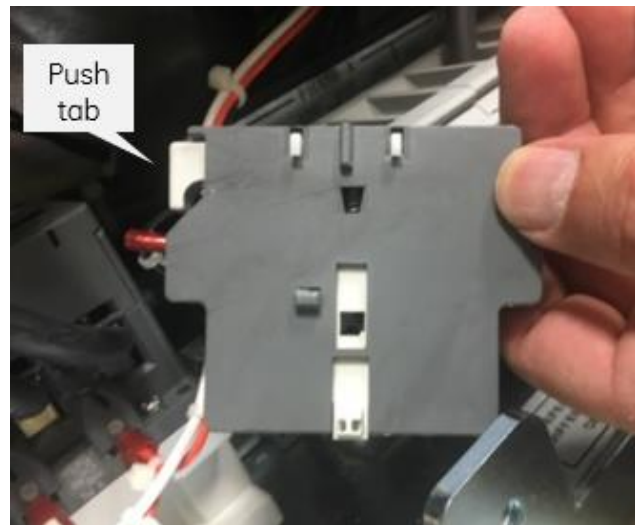




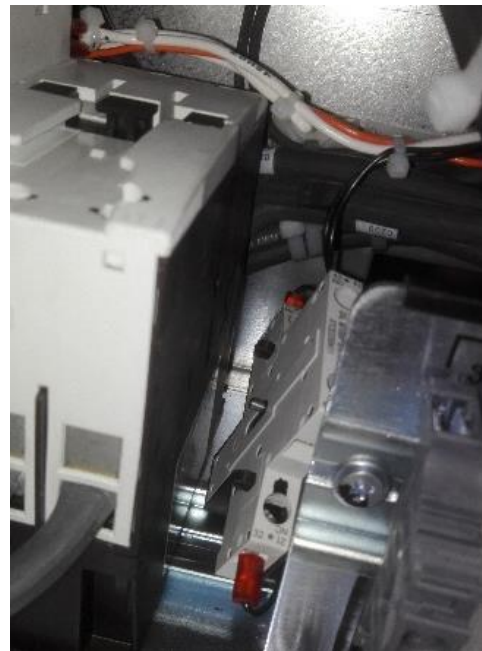
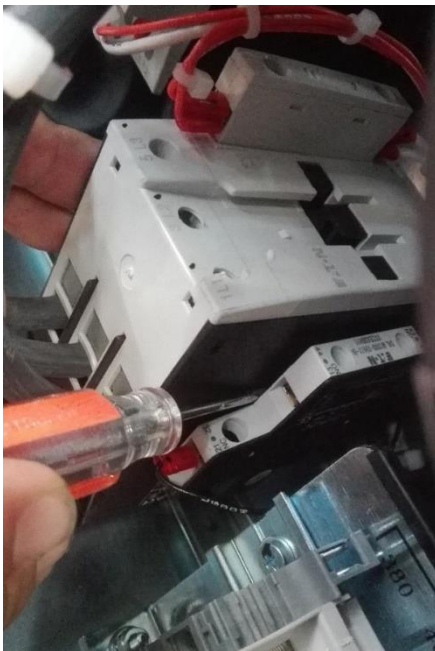
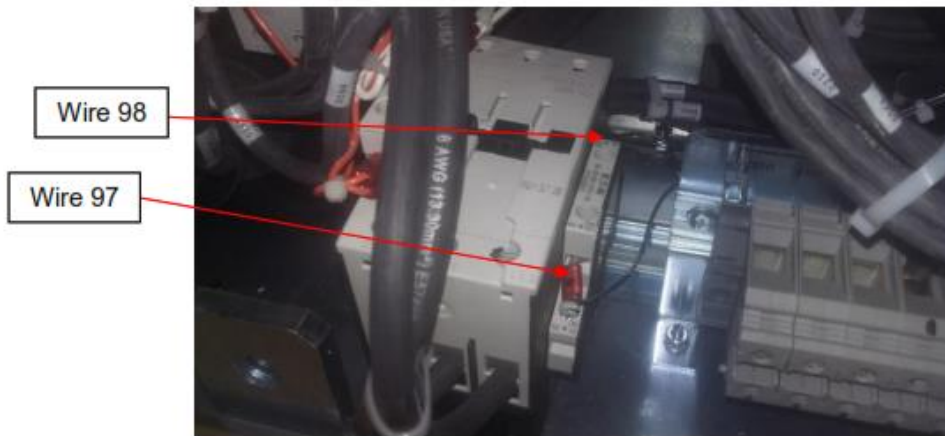
New Replacement Step when replacing Condor with Delta for the 1st time. There is a difference between CON1 mounting for XGD (mounted vertically) and XGD2 (mounted horizontally). There's no need to remove CON1, the AUX switch can be removed to change from NC to NO contacts.

With the Delta power supply assembly installed, the auxiliary contact wires (97 and 98) on CON1 contactor must be moved from the (NC) Normally Closed to (NO) Normally Open location. The terminals used on the AUX wires are spring brass fork terminals, the screws will not need to be removed. Tighten the screws in terminals 21 and 22 once the wires have been moved as they will no longer be in use.

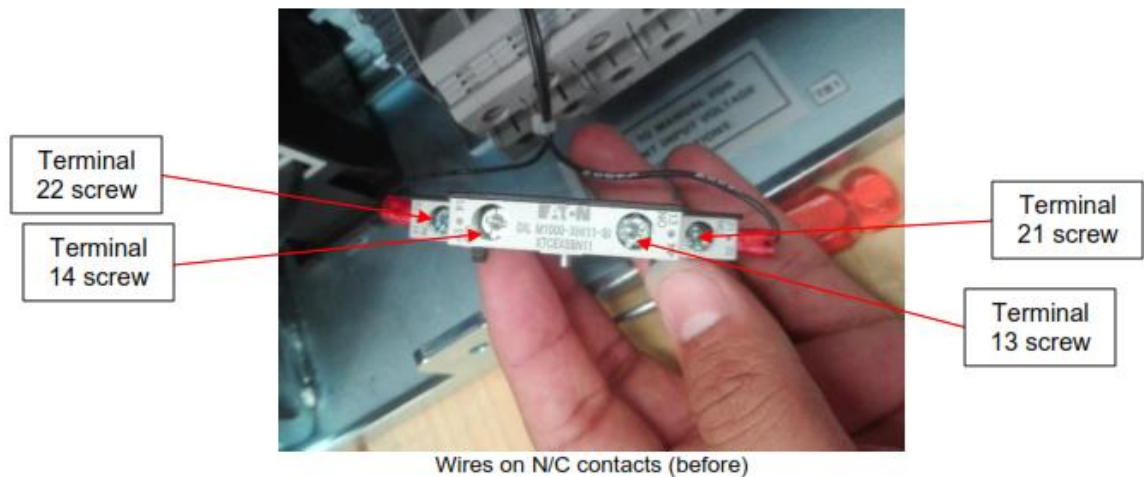
1. For CON1 AUX SW on the XGD, the auxiliary switch does not need to be removed, the auxiliary contact terminals can be accessed from the front. If you need more room to work, gently push in the light gray tab on the base of the AUX SW, this will release auxiliary switch from CON1.



2. Use a Philips screwdriver to disconnect wire 97 from the N/C contact, terminal 21 and remove the wire to the N/O contact, terminal 13. Do the same for wire 98, moving it from the N/C contact, terminal 22 to the N/O contact, terminal 14.
3. Push the light gray tab to reinsert AUX SW onto CON1.
4. For CON1 AUX SW on the XGD2, insert a small flat-head screwdriver between contactor CON1 and the auxiliary contact. Gently pry the auxiliary contact away from the contactor.

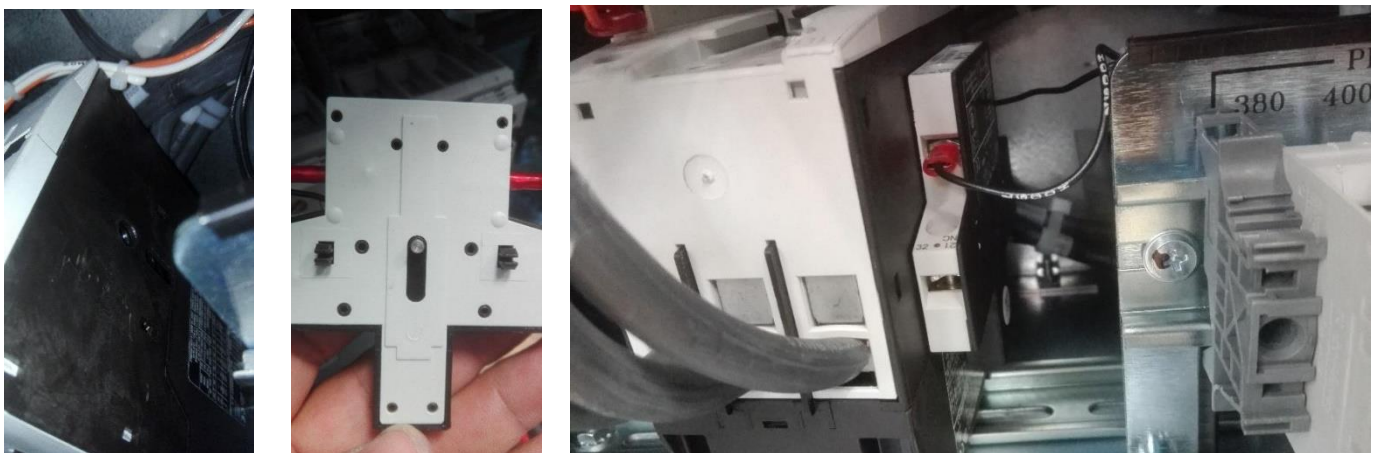


5. Use a Philips screwdriver to disconnect wire 97 from the N/C contact, terminal 21 and remove the wire to the N/O contact, terminal 13. Do the same for wire 98, moving it from the N/C contact, terminal 22 to the N/O contact, terminal 14.



6. Once the wires have been moved, push the auxiliary contact to the contactor, CON1 when the peg and clips of the auxiliary contact are aligned.

Note: An audible “click” will be heard when the auxiliary contact is fully inserted back onto the contactor.



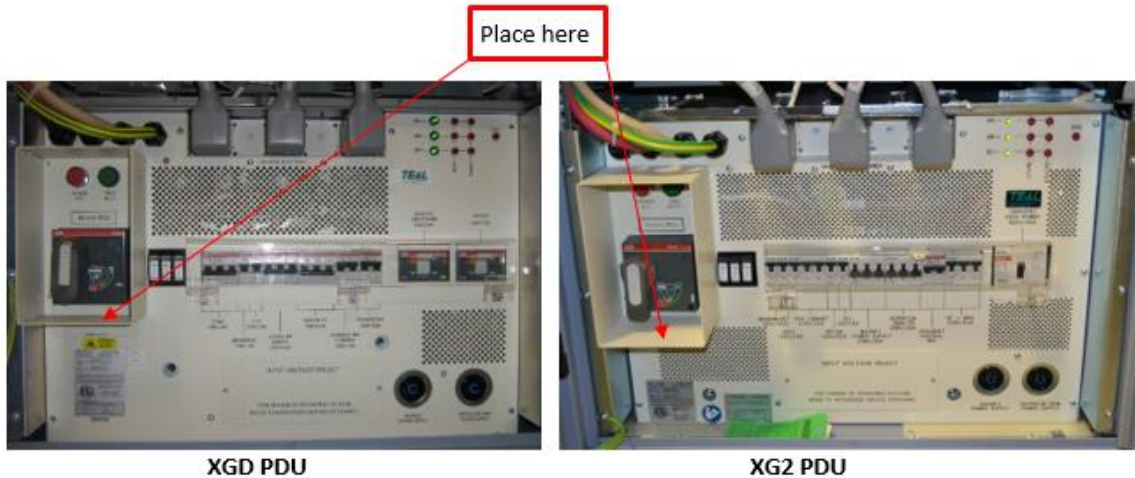
Note: Before you reinstall the front panel, make sure you didn’t bump the toggle switch on the control board, located above the power supplies. For XGD the switch is left (LED is green) and for XGD2 the switch is right (LED is yellow).

7. Reinstall the front panel.

Note: Be careful not to damage the leak sensor strip.

8. Apply maintenance sticker (5661793) noting that the Delta supplies are installed on the front of the PDU, with the following information:

Date _____
 Component This PDU is equipped with Delta 48V PS (5745311)
 Employee ID _____



Finalization

1. Remove LOTO at the MDP, power up the PDU, and perform *Check Scan*.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
A0			New Release
A1	5/28/10	R. Gregg	1. Changed "5-20" to "NEMA 5-20R" in section 2.1 2. Added TEAL FRU P/N's to table 4-1 3. Added sentence "The illustration shows TB1 configured for 380V Delta" in section 3.1.1 4. Corrected "PS3" to be "PS2" in section 4.1.6 title 5. Change "board" to "Control Board" and "Power Supply" to "PS1 Power Supply" in table 4-2 6. Removed the cause "Open from J3-2 to J3-3" and its respective solutions from table 4-2, troubleshooting guide 7. Removed solutions 1-3 from cause "Defective Control Board A1" in table 4-2 troubleshooting guide 8. Rewrote the solution section of cause "PS1 Power Supply not receiving power" in table 4-2 troubleshooting guide 9. Added Symptom of "Internal system AC line clock signal missing." Add Cause of "CB12 open.". Add Solution of "Reset CB12 "CAM Sync Reset" from PDU front panel." to table 4-2 troubleshooting guide 10. Replaced component location illustrations, 3-5 & 3-6 with correct version drawings 11. Replaced "XGD" with "XGD2" in sections 1.1, 1.2, 2.3 and various headers 12. Added reference to CB12 in table in section 2.1

			13. Corrected 4 th instruction in section 4.1.3 to be numbered “4.” instead of “1.”
A2	6/28/10	GEHC	Alternate torque values added by GEHC to section 4.1.6
A3	6/7/12	R. Gregg	1. Added photos and explanation of control board switch configuration to section 4.1.3 2. Added photo of LED power supply LED and explanation of LED purpose to section 4.1.6 3. Changed text in section 3.1.1.4 to read 480V 4. Added text “and the PDU is not in E-Stop mode” to section 4.1.6, 2nd paragraph. 5. Added new cause and solutions to troubleshooting table 4-2 in section 4.1.8 6. Removed reference to 5343114-3 PEM configured control board, added 5343114-5 to table 4-1 in section 4 7. Added power supply troubleshooting to table 4-2 in section 4.1.8
A4	3/13/13	D. Bilak	Added 60601-1 3 rd edition safety comments.
A5	3/21/14	T. Donaldson	1. Corrected callouts for the fan connectors in section 4.1.5 2. Added section 4.1.5.1 and 4.1.5.2.- Replacement of Fans (FAN1 or FAN2)
A6	10/15/15	E. Tamayo	1. Added section 3.3 2. Added section 7 to 4.1.3 to clarify seating location of 15 pin connector cable bundle prior to closing. 3. Removed callouts for previously removed CB12 5. Group images in Section 4.1 to make it more clear.
07	10/4/18	S. Woods	1. Updated PS FRU kit part number 2. Updated PS FRU kit installation instructions 3. Updated formatting 4. Replaced schematic diagram with new diagram
08	10/2/19	S. Woods	1. Added FRU kit instructions for 4991422, 4991423, 4991424, 4991425, 4991426, 4991427, 4991437, 4991438, 4991525, 4991526, 4991528, and 4991529 2. Updated how to snap off contactors from din rail 3. Added disclaimer notes for some FRUs 4. Added description of MTE part number and revision 5. Updated FRU list part numbers 6. Added Appendix A
09	12/9/19	S. Woods	1. Removed FRU 4991425 and corresponding installation instructions